**AFRL-AFOSR-VA-TR-2025-0503**

# Thermo-Hydrodynamics of a Strongly Coupled Plasma

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**Final Technical Report**

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**14. ABSTRACT**
Final report summarizing a project focused on strongly coupled plasmas. Excellent imaging results leveraging compressed ultrafast photography in collaboration with Prof. Lihong Wang at Cal Tech. The report discusses several high impact pubs that came from the grant.


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FA9550-21-1-0295

AFOSR Final Report on the Project

Thermo-Hydrodynamics of a Strongly Coupled Plasma

Reporting Period 7/1/2021 - 4/15/2025

Approved for Public Release

Submitted to Program Managers: Dr Joel Bixler; Dr Andrew Stickrath and Dr John Luginsland

Principal Investigator: Seth Putterman; puherman@ritva.physics.ucla.edu

Thermo-Hydrodynamics of a Strongly Coupled Plasma

ABSTRACT

Plasmas are the ultimate state of energized matter. The goal of this project is to carry out experiments and theory aimed at understanding plasmas with a high density such as collisional plasmas and strongly coupled plasmas [SCP] where the coulomb interaction dominates thermal motion.  Dense plasmas, form on hypersonic vehicles and furthermore appear in inertial confinement fusion and laser matter interaction. They characterize molten salts and are of keen importance to astrophysics. Experimental probes of the plasmas in a laboratory environment are limited and furthermore until now theoretical insights require the use of high cost particle simulations. This project has successfully developed laboratory experimental probes of these plasma via the application of calibrated - compressed ultrafast photography [500Bfps] to laser breakdown of dense gases. A theoretical approach which generalizes Navier- Stokes hydrodynamics to nonlocal stresses has been developed which easily reproduces simulations of the various dispersion laws and furthermore extends our capabilities to the nonlinear domain. We have also demonstrated a spherical force field in a collisional microwave generated plasma. This last experiment makes it possible for the first time to validate theories of spherical convection in ground based laboratories. We have measured recombination within the dense plasmas in the parameter space that forms on hypersonic vehicles. This is a key process that controls their turbulence and affects communication. In summary we have delivered: A) first Navier-Stokes level hydrodynamics of a strongly coupled plasma, B) first reproducible hardware experiments that produce and photograph in 2D and with 2ps resolution the creation and evolution of strongly coupled real plasmas, C) the first laboratory system to produce convection in spherical 'gravity' which is important to theories of space weather.

FA9550-21-1-0295

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Introduction to Final Technical Report:

A key parameter in describing a plasma is the coupling coefficient $\Gamma = e^2/akT$ where '$e$' is the fundamental unit of electric charge '$a$' is the inter ion spacing, $k$ is Boltzmann's constant and $T$ the temperature. Dilute plasmas such as studied in Stellarators and Tokamaks have $\Gamma \ll 1$ whereas a strongly coupled plasma [SCP] has $\Gamma \sim > 1$ . This is the limit where electrostatic interactions compete with or dominate thermally driven motion. Strongly interacting plasmas are characterized by their collisionality as well as the strength of the electrostatic interactions. In addition to $\Gamma$ there is now a kinetic collision time $\tau$, which leads to the dimensionless quantity "$\omega\tau$" where $\omega$ is a dynamic frequency such as appears in a dispersion law. A plasma that is sufficiently dense so as to be collisional has $\omega\tau \ll 1$. The plasmas which are dense enough to be collisional or strongly interacting play a role in solar dynamics. These plasmas also form on hypersonic vehicles. In addition dense plasmas characterize stages of inertial confinement fusion. The purpose of this project was to develop: I ) a theory of these systems and II) laboratory based experiments which generate and measure these systems.
Here is a summary of achievements:

1) We have achieved a unified theory of a strongly coupled plasma [$\Gamma \gg 1$] at the level of Navier-Stokes hydrodynamics. A key paper which we view as opening up a new field has been published in the Physics of Fluids https://doi.org/10.1063/5.0194352
"Variational principles for the hydrodynamics of the classical one-component plasma"
We view our advances as replacing kinetic theory approaches and for the first time opening up the study of nonlinear processes in these systems. I recommend to read the introduction of the above extensive paper and also to look at the Overview in this report that is entitled:
• Extending the Limits of Hydrodynamics to Dense Plasmas

The key point is that we have a simple unified theory that recovers in detail the longitudinal and transverse dispersion laws as well as matching the high frequency sum rules. Amazingly this theory works down to distance *shorter* than the inter-particle spacing. So its range of applicability is larger than ordinary hydrodynamics. This is a bonus of the strong coupling limit: where the coulomb forces dominate the random thermal motion.

Looking ahead this theory can be extended to a Yukawa potential and to a 2 component plasma. A wealth of nonlinear processes ; such as shock waves can now be self-consistently calculated for a strongly coupled plasma and even for molten salts for the first time.

2) This project is unique in that it includes both fundamental theory and experiment on a system which is of compelling interest to the frontiers of plasma physics namely dense plasmas.

Experiments to generate and measure an SCP were carried out in collaboration with the Lab of Lihong Wang at Caltech. For each laser breakdown event in a dense gas we have succeeded to take a 2D frame every 2ps = 500Bfps. A paper with example movies has been posted on the Arxiv and is available at this url: http://arxiv.org/abs/2507.04608

"Compressed Ultrafast Photography of Plasmas Formed from Laser Breakdown of Dense Gases Reveals that Internal Processes Dominate Evolution at Early Times"

Here is a summary of the first 4 figures which provide unique experimental data for this field:

Fig.1 showing 2D images from a movie of breakdown in 2bar xenon where the length of the movie is .5ns and there are 400frames. One can see the evolution of the plasma shape.

Fig. 2 calibrated dynamic intensity of light as see through red, green and blue filters

Fig. 3 development of breakdown on a ps timescale as a function of position

Fig. 4 Time evolution of contours of constant emission showing that at short time the breakdown region of the highest pressure gases contracts. This is possibly due to the tensile strength that has its origin in the nature of the SCP.

From the spectra one obtains emissivity and temperature which in turn lead to the degree of ionization as plotted for example in Figure 9. The ionization strongly disagrees with accepted theories as applied to this region of parameter space. Yet note that there are no experiments which differ from our findings. An evaluation of transport properties indicates that during the first 10ns the plasma is an isolated system evolving according to its own dynamics with little interaction with the environment. Thus measurements of the plasma during this period will enable measurements, as compared to simulations, of the equations of state. Strongly coupled plasmas are of compelling interest and these are the first 2D measurements. To our knowledge this is the first time that compressed ultrafast photography has been used to uncover new physics. The paper, for the above url is appended to this report.

3) We have used pulsed microwaves to excite sound in a plasma that is sufficiently dense that it is collisional. The nonlinear stresses are sufficiently strong that this plasma stratifies and develops an effective spherical gravity which then drives a spherical convection. A laboratory system displaying spherical convection has been a long standing goal of solar research. The effective value of gravity '$g_{eff}$' that could previously be achieved was so small that it was swamped by earth's vertical 'g'. As a result these early experiments were flown in the microgravity environment of space. Our dense plasma system achieved $[g_{eff}/g]>100$. This paper

received an Alt Metric [Am] score of 445 placing well into the top 1% of papers published in Physical Review Letters.
"Thermal Convection in a Central Force Field Mediated by Sound"
https://journals.aps.org/prl/abstract/10.1103/PhysRevLett.130.034002
This paper was an Editor's Suggestion and Featured in Physics and reported by Wired Magazine Feb 20, 2023. A major aspect of the excitement generated by this work lies on our uncovering 2 terms in the Navier –Stokes hydrodynamics that were overlooked in previous theory of radiation pressure.

4) Our ground based steady state hypersonic mimetic plasma is amenable to the study of its equations of state and transport properties. This is a unique capability. A key parameter that is essential to chemistry under these stressed conditions is the recombination time. We have developed a unique method which uses the in phase and out of phase [I&Q] reflected signal to determine charge and temperature. This provides an unusually clean comparison with various theories such as the famous 3 body recombination effect. The below figure encapsulates the key result; which is that the approach of plasma charge to local thermodynamic equilibrium lags the Change in temperature:

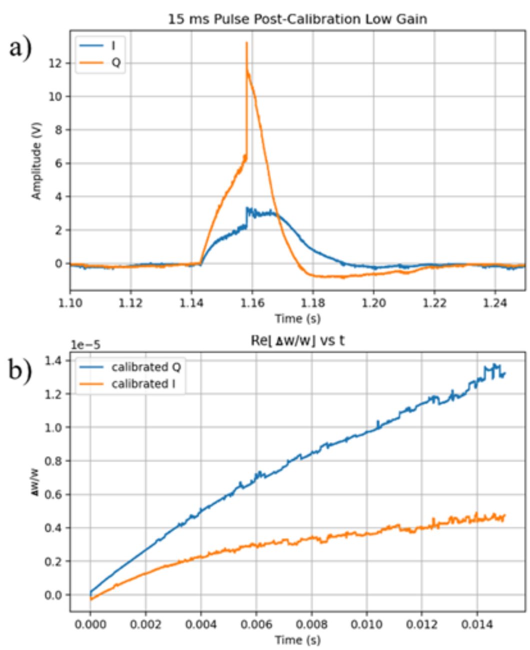



Figure showing I, Q after there is an interruption to the power driving the plasma. The power is turned off at 1.143ms and turned back on 15ms later. In Figure 'a' the orange Q curve is a proxy for temperature and the blue curve is a proxy for charge. Notice that when the power is turned back on the Q curve first jumps because the signal has increased but then drops immediately.as temperature closely follows power on the ms time scale. After showing a jump due to an increased signal the blue curve hangs around for about 10ms. This is the recombination time. A report describing this experiment is appended.

Here are future opportunities regarding our project on microwave generated plasmas [3,4] above] : a) the chemistry of a plasma which forms on the next generation hypersonic vehicles which will reach 10,000K is an unknown and compelling issue which can be set up and measured with our apparatus. Figure 9.12 in Anderson [Hypersonic and High Temperature Gas Dynamics] is key and yet has not been subject to experimental testing; b) our system should become a plasma oscillator when the drive is sufficiently strong so as to reach 10,000K . In this case a continuous input of microwave energy will cause the gas to oscillate and also pull in from the walls and confine itself.

# Overview





**Why hydrodynamics?**

Hydrodynamics is an effective description in terms of a few macroscopic variables.

Successful for many systems and problems, such as:

- gas, liquid, solid, plasma,
- linear and nonlinear problems.

As it is an effective description, there is a need to explore its limits, such as at small length scales.

# Overview

**Exploring limits of hydrodynamics**

**Strongly coupled one-component plasma**

- Derived generalized hydrodynamic equations that extend Navier-Stokes equations
- Used variational (least action) principle that includes nonlocal terms
- Compared dispersion laws to numerical data

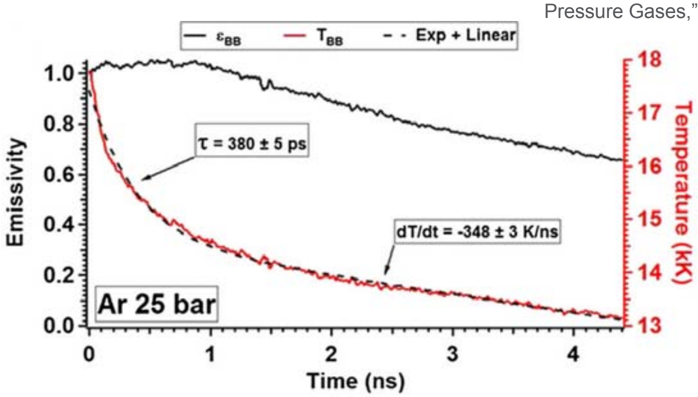
**Extreme bubble collapse in compressible fluids**

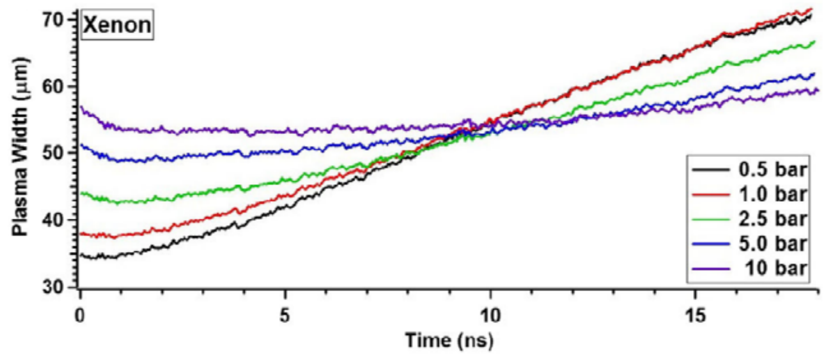
- Developed a numerical solver allowing for quick and accurate computations
- Computed an analytic benchmark testing the accuracy of solvers
- Solved the problem for experimentally relevant parameters and typical fluids

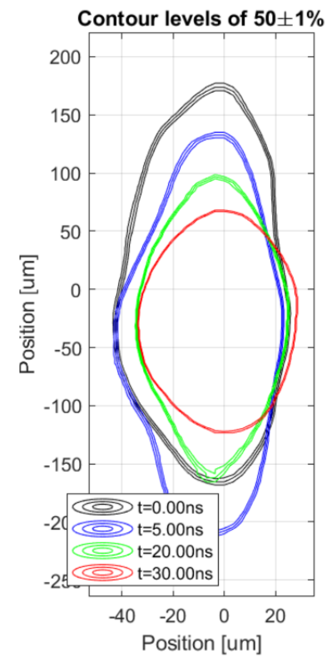
# Strongly coupled plasma: experiments



A. Bataller, G. R. Plateau, B. Kappus, and S. Putterman, “Blackbody Emission from Laser Breakdown in High-Pressure Gases,” Phys. Rev. Lett. 113, 075001 (2014)




Xenon, 21.5 bar

unpublished data from collaborators at Caltech, data taken by Peng Wang in the laboratory of Lihong Wang, analyzed by Seth Pree and John Koulakis

$$\Gamma = \frac{q^2}{4\pi\varepsilon_0 a} \Big/ k_B T \qquad N\left(\frac{4\pi a^3}{3}\right) = V$$

# Strongly coupled plasma: theoretical approaches

**Modeling strongly coupled plasmas**

**Molecular dynamics**
**Issue: nonlinear problems**

**Kinetic theory**
**Issue: closure, thermodynamics**

**Hydrodynamics**

"(..) we conclude that (..) the hydrodynamic description in the manner of Navier and Stokes is not valid for the OCP at any coupling strength $\Gamma$."

J. P. Mithen, J. Daligault, and G. Gregori, "Extent of validity of the hydrodynamic description of ions in dense plasmas," Phys. Rev. E 83, 015401(R) (2011)

The range of validity of hydrodynamics for strongly coupled plasmas is greater than for ordinary fluids and works up to length scales comparable to $a$.

**Variational principles for the hydrodynamics of the classical one-component plasma**

Cite as: Phys. Fluids **36**, 037131 (2024); doi: 10.1063/5.0194352
Submitted: 26 December 2023 · Accepted: 18 February 2024 ·
Published Online: 12 March 2024

Daniels Krimans[a] and Seth Putterman

# Strongly coupled plasma: equilibrium properties

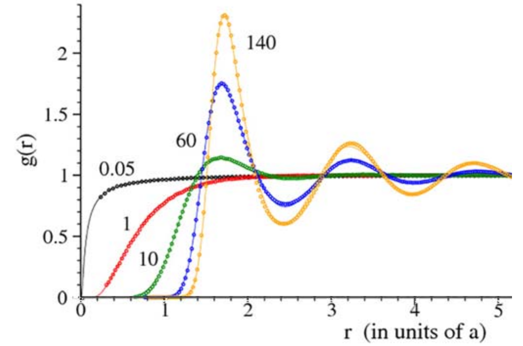
In thermal equilibrium:

$$E = \frac{3}{2}Nk_BT + \frac{N^2}{2V}\int \underline{g\left(n,T,|\vec{r}|\right)}\,\phi\left(|\vec{r}|\right)\mathrm{d}\vec{r}$$

$$= Nk_BT\left(\frac{3}{2} + \underline{u(\Gamma)}\right)$$

$$u(\Gamma) = -0.9\Gamma + 0.5944\Gamma^{1/3} - 0.2786$$

S. A. Khrapak and A. G. Khrapak, "Simple thermodynamics of strongly coupled one-component-plasma in two and three dimensions," Phys. Plasmas 21, 104505 (2014)




N. Desbiens, P. Arnault, and J. Clerouin, "Parametrization of pair correlation function and static structure factor of the one component plasma across coupling regimes," Phys. Plasmas 23, 092120 (2016)

Our variational approach:

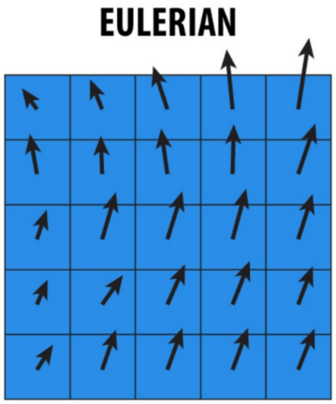
$$L = \int \underbrace{\left(\frac{1}{2}mn\vec{v}^2 - \frac{3}{2}nk_BT\right)}_{\text{ideal (Euler's) hydrodynamics}}\mathrm{d}\vec{x} - \iint \underbrace{\frac{nn'}{2}g\left(n,T,|\vec{x}-\vec{x}'|\right)\phi\left(|\vec{x}-\vec{x}'|\right)\mathrm{d}\vec{x}\mathrm{d}\vec{x}'}_{\text{nonlocal terms for strong coupling}}$$

ideal (Euler's) hydrodynamics

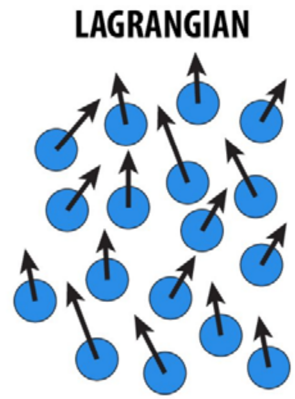
nonlocal terms for strong coupling

# Strongly coupled plasma: Eulerian vs Lagrangian






Tracks properties at fixed grid locations given by positions $\vec{x}$.




Tracks properties of moving particles starting from initial positions $\vec{x}_{\text{init}}$.

Out-of-equilibrium behavior of $g\left(n, T, |\vec{x} - \vec{x}'|\right)$?

| | | |
|---|---|---|
| $g(n, T, \|\vec{x} - \vec{x}'\|)$ | $g(n, T, \|\vec{x}_{\text{init}} - \vec{x}'_{\text{init}}\|)$ | $g(n_{\text{init}}, T, \|\vec{x}_{\text{init}} - \vec{x}'_{\text{init}}\|)$ |
| Eulerian | Lagrangian | Modified Lagrangian |



http://15462.courses.cs.cmu.edu/fall2018/lecture/pdes/slide_023, Carnegie Mellon University (2018)

# Strongly coupled plasma: equations of motion

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Euler's equation:

$$mn\left(\frac{\partial\vec{v}}{\partial t}+\left(\vec{v}\cdot\vec{\nabla}\right)\vec{v}\right)=-\vec{\nabla}\left(\frac{3}{2}k_Bn^2\frac{\partial T}{\partial n}\Big|_s\right)$$

OCP Eulerian approach:

$$mn\left(\frac{\partial\vec{v}}{\partial t}+\left(\vec{v}\cdot\vec{\nabla}\right)\vec{v}\right)=-\vec{\nabla}\left(\frac{3}{2}k_Bn^2\frac{\partial T}{\partial n}\Big|_s+\frac{1}{2}\int n^2n'\frac{\partial g}{\partial n}\Big|_s\frac{q^2}{4\pi\varepsilon_0|\vec{x}-\vec{x}'|}\mathrm{d}\vec{x}'\right)$$

$$\underline{-n\vec{\nabla}\int n'\left(\frac{g+g^T}{2}\right)\frac{q^2}{4\pi\varepsilon_0|\vec{x}-\vec{x}'|}\mathrm{d}\vec{x}'}+n\vec{\nabla}\int n'_-\frac{q^2}{4\pi\varepsilon_0|\vec{x}-\vec{x}'|}\mathrm{d}\vec{x}'$$

OCP Lagrangian approach:

$$mn\left(\frac{\partial\vec{v}}{\partial t}+\left(\vec{v}\cdot\vec{\nabla}\right)\vec{v}\right)=-\vec{\nabla}\left(\frac{3}{2}k_Bn^2\frac{\partial T}{\partial n}\Big|_s+\frac{1}{2}\int n^2n'\underline{\frac{\partial g}{\partial n}}\Big|_s\frac{q^2}{4\pi\varepsilon_0|\vec{x}-\vec{x}'|}\mathrm{d}\vec{x}'\right)$$

$$\underline{-n\int n'\left(\frac{g+g^T}{2}\right)\vec{\nabla}\left(\frac{q^2}{4\pi\varepsilon_0|\vec{x}-\vec{x}'|}\right)\mathrm{d}\vec{x}'}+n\vec{\nabla}\int n'_-\frac{q^2}{4\pi\varepsilon_0|\vec{x}-\vec{x}'|}\mathrm{d}\vec{x}'$$

$$\vec{\nabla}\times\int n'\left(\frac{g+g^T}{2}\right)\vec{\nabla}\left(\frac{q^2}{4\pi\varepsilon|\vec{x}-\vec{x}'|}\right)\mathrm{d}\vec{x}'\neq\vec{0}$$

# Strongly coupled plasma: transverse modes



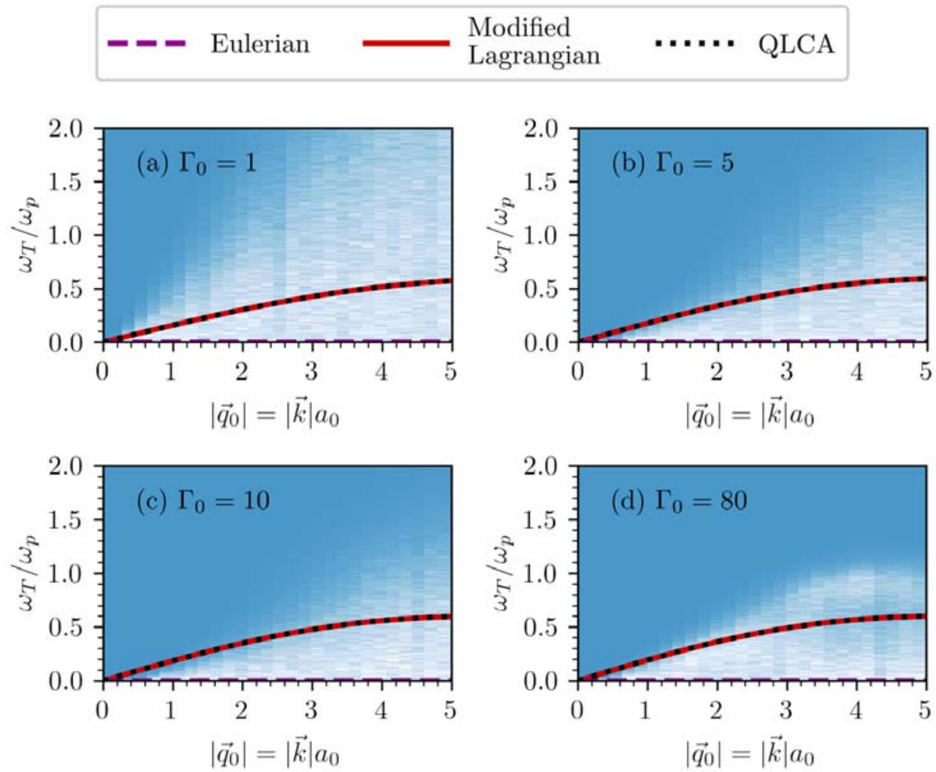
OCP Eulerian approach: $\omega_T = 0$

OCP Lagrangian and Modified Lagrangian approach:

$$\omega_T^2(|\vec{k}|) = \frac{q_+^2 n_0}{m\varepsilon_0} \int_0^\infty \frac{(g-1)_0}{|\vec{k}|r^2} \left( \sin(|\vec{k}|r) + 3\frac{\cos(|\vec{k}|r)}{|\vec{k}|r} - 3\frac{\sin(|\vec{k}|r)}{|\vec{k}|^2 r^2} \right) \mathrm{d}r$$

QLCA (Quasilocalized charge approximation):

- not consistent with thermodynamics,
- cannot be applied to nonlinear problems.

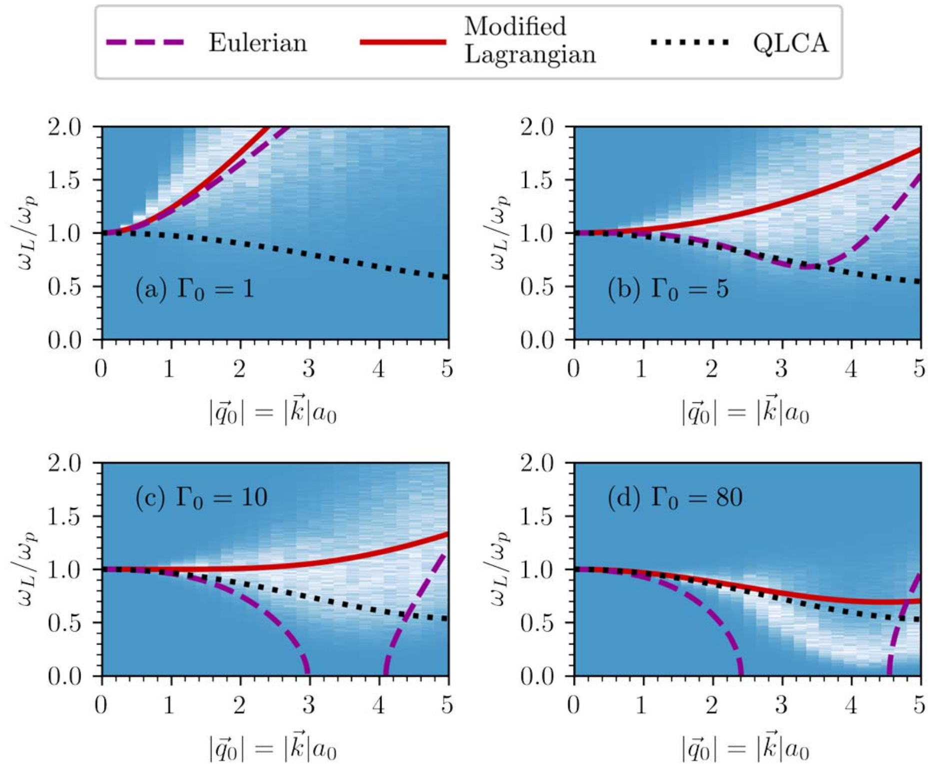


# Strongly coupled plasma: longitudinal modes



OCP Modified Lagrangian approach and QLCA:

$$
\begin{aligned}
\omega_L^2(|\vec{k}|) = \underline{\frac{q_+^2 n_0}{m\varepsilon_0}} + |\vec{k}|^2 \Bigg( & \left[\frac{\partial}{\partial n}\left(\frac{3k_B}{2m} n^2 \frac{\partial T}{\partial n}\Big|_s\right)\Big|_s\right]_0 \\
& + \frac{q_+^2 n_0}{2m\varepsilon_0}\int_0^\infty r\left[\frac{\partial}{\partial n}\left(n^2\frac{\partial (g-1)}{\partial n}\Big|_{s,r}\right)\Big|_{s,r}\right]_0 \mathrm{d}r \Bigg) \\
& + \frac{q_+^2 n_0^2}{m\varepsilon_0}\int_0^\infty \frac{1}{r}\left[\frac{\partial (g-1)}{\partial n}\Big|_{s,r}\right]_0 \left(\frac{\sin(|\vec{k}|r)}{|\vec{k}|r} - \cos(|\vec{k}|r)\right) \mathrm{d}r \\
& \underline{+ \frac{2q_+^2 n_0}{m\varepsilon_0}\int_0^\infty \frac{(g-1)_0}{|\vec{k}|r^2}\left(3\frac{\sin(|\vec{k}|r)}{|\vec{k}|^2 r^2} - 3\frac{\cos(|\vec{k}|r)}{|\vec{k}|r} - \sin(|\vec{k}|r)\right) \mathrm{d}r}
\end{aligned}
$$




Hydrodynamic description is accurate on scales $\lambda \approx a$.

# Strongly coupled plasma: ideas for the future

**More complicated nonlinear problems, such as:**

- plasma expansion,
- motion of vortices,
- shock waves.

**More complicated systems, such as:**

- Yukawa fluids,
- two-component plasma,
- warm dense matter,
- systems consisting of electric or magnetic dipoles.

# Compressed Ultrafast Photography of Plasmas Formed from Laser Breakdown of Dense Gases Reveals that Internal Processes Dominate Evolution at Early Times

Peng Wang[1], Yogeshwar Nath Mishra[1,2,3], Seth Pree[4], Lihong V. Wang[1,*],
Dag Hanstorp[5], John P. Koulakis[6], Daniels Krimans[6], and Seth Putterman[6,*]
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(Dated: July 8, 2025)

Compressed ultrafast photography (CUP) is applied to laser breakdown in argon and xenon under pressures up to 40 bar to obtain 2D images of the plasma dynamics of single events with a spatial resolution of $250 \times 100$ pixels and an equivalent frame rate of 500 GHz. Light emission as a function of position and time is measured through red, green, blue, and broad-band filters. The spatially encoded and temporally sheared image normally used in CUP is now enhanced by the introduction of a constraint given by a spatially integrated and temporally sheared unencoded signal. The data yield insights into the temperature, opacity, the plasma formation process, and heat flow within the plasma and to the surrounding ambient gas. Contours of constant emission indicate that plasmas formed from sufficiently dense gas contract rather than expand despite having a temperature of a few eV. Plasmas formed from relatively low pressure gases such as 7 bar argon can radiate with emissivity near unity. Modeling transport and opacity as arising from inverse Bremsstrahlung requires a degree of ionization that strongly exceeds expectations based on Saha's equation even as customarily modified to include density and screening. According to this model, both electrons and ions are strongly coupled with a plasma coefficient $\gtrsim 1$. During the first few nanoseconds after formation, Stefan-Boltzmann radiation and thermal conduction to ambient gas are too weak to explain the observed cooling rates, suggesting that transport within the plasma dominates its evolution. Yet, thermal conduction within the plasma itself is also small as indicated by the persistence of thermal inhomogeneities for far longer timescales. The fact that plasma is isolated from the surroundings makes it an excellent system for the study of the equation of state and hydrodynamics of such dense plasmas via the systems and techniques described.

In a strongly coupled plasma (SCP) the electrostatic potential energy is comparable to or greater than the thermal kinetic energy so that the fundamental plasma parameter $\Gamma = e^2/akT \gtrsim 1$, where $e$ is the fundamental unit of charge, $T$ is the temperature, $ax^{1/3} = a_0 = (3/4\pi n_0)^{1/3}$, $n_0$ is the atomic density and $x \leq 1$ is the degree of ionization. This regime is important for the study of astrophysical objects (e.g., white dwarf stars [1, 2]), dusty plasmas [3], ultracold atom/ion trapping [4, 5], ultrafast laser breakdown [6, 7], photoelectron sources [8] and inertial confinement fusion [9]. It should be contrasted with, say, Tokamak plasmas where the density is low and the temperature is high so that $\Gamma \ll 1$ and where the constituent particles are weakly interacting similar to those in an ideal gas. On the other hand, for example, warm dense matter is a strongly coupled plasma which is described by Saumon et al. [10] as a regime where "all the physics is important."Despite numerous studies, controlling and measuring strongly coupled plasmas is a challenge that is here addressed with the application of Compressed Ultrafast Photography (CUP, [11, 12]) to the laser breakdown of a dense gas which we believe shows strong coupling. Data generated by this technique show that on the nanosecond time scale the breakdown plasma is only weakly coupled to the environment so that its intrinsic thermohydrodynamic properties can be probed.

Due to the long-range nature of the Coulomb forces and the resulting correlation between particles, the macroscopic response of a SCP is also expected to be long-ranged [13–16] as compared to Navier-Stokes hydrodynamics. So a line of attack for investigating SCP is to image their macroscopic motions. This is achieved by using CUP to capture the time-resolved, two-dimensional evolution of a SCP that is created by laser breakdown of dense argon and xenon. Example data that can be obtained with this technique are shown in Figures 1-4. Figure 1 shows the evolution of a plasma formed from laser breakdown of 2 bar xenon as measured through a filter centered on 620 nm. Breakdown is driven by a focused 0.9 mJ, 220 fs long pulse of 800 nm light using

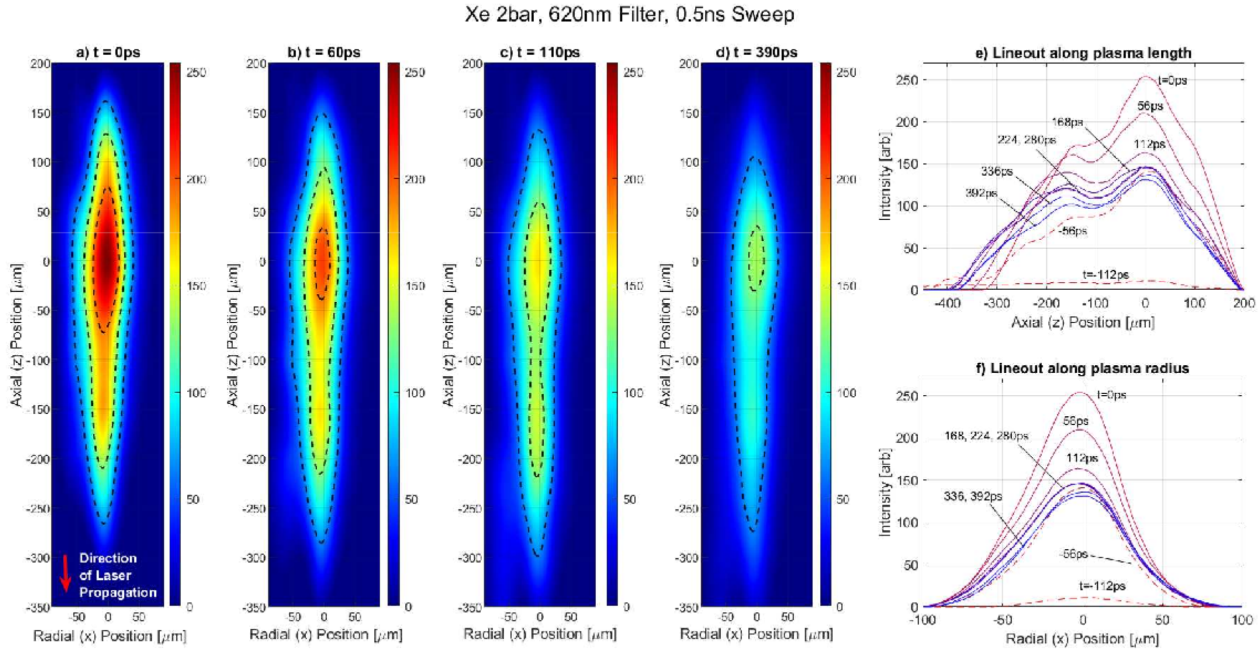


FIG. 1. a)-d) Four frames of the breakdown of 2 bar xenon with a CUP sweep setting of 0.5 ns. Extracted frames are at t=0 (peak emission), 60, 110 and 390 ps. A single-shot movie has 400 frames. The intensity is false colored and in arbitrary units. Topographic lines of constant intensity are at 25%, 50%, and 75% of the peak intensity of each frame. e), f) Lineouts of the axial $z$ and radial $x$-dependence of the emission during the first 400 ps. The lines cross the region of maximum emission intensity. This data is from a single event. The fact that CUP has accurately captured multiple peaks in the axial direction is confirmed by traditional streak photos (Supplemental Material fig S7).

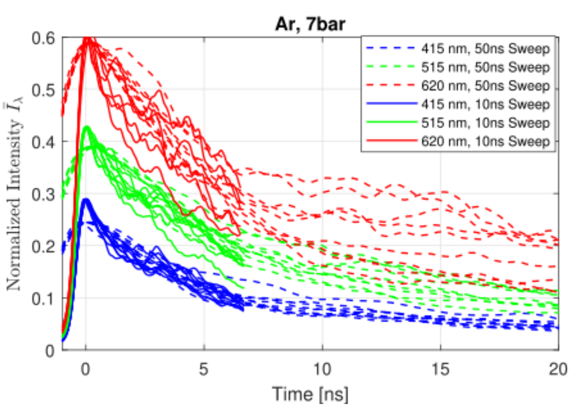


FIG. 2. The measured emission (units of MW m$^{-2}$ nm$^{-1}$ per steradian) from the brightest 5x5 pixels region has been scaled with $\lambda^5/2hc^2$ to yield a dimensionless emission. Each trace is a single shot through Red (620 nm), Green (515 nm), and Blue (415 nm) filters for 10 ns and 50 ns sweeps.

a windowed pressure cell [7]. Figure 2 shows the intensity of emission from the bright central region of breakdown of 7 bar argon as taken through 620 nm ("Red"), 515 nm ("Green"), and 415 nm ("Blue") filters. The plotted emission intensity per steradian has been scaled to $\bar{I}_\lambda = I_\lambda \lambda^5/2hc^2$. Figure 3 shows the rise of the plasma emission for different positions along the plasma length. Since the curves at various positions have the same shape, we interpret the measured rise time as being limited by the system temporal resolution. None-the-less, the time delay between curves at different positions gives information about the plasma formation processes. Figure 4 shows the evolution of contours of constant emission that are normalized to the instantaneous peak intensity for 7 and 25 bar argon. These frames and intensity curves come from single shot data acquisitions and are not the result of patching together separate pump/probe acquisitions. The equivalent frame rate can reach 500 billion fps though we estimate the resolution to be 30 ps.

Analysis of the spectral components indicates that argon and xenon at pressures above 7 bar emit with emissivity close to unity and are opaque. At these atomic densities, the experimentally measured opacity requires a degree of ionization $x \gtrsim 1$ assuming that opacity arises from inverse Bremsstrahlung. Combined with our measurement of temperature, we find that $\Gamma$ is order unity for both electrons and ions. These systems can therefore be characterized as strongly coupled plasmas.

The goal of this communication is to describe the experiment which acquires the above data and the theoretical analysis which leads us to conclude that,

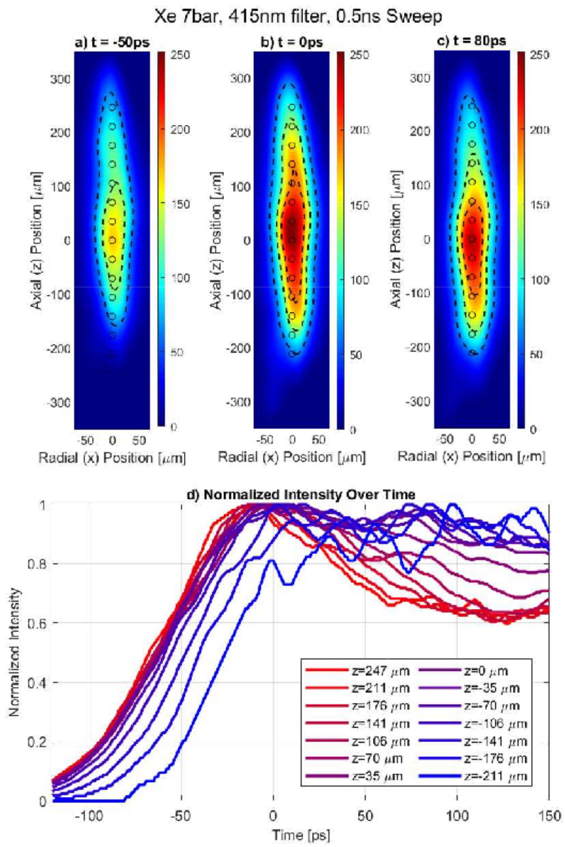


FIG. 3. Rise of plasma emission as a function of position. Panels a), b), and c) are frames at $t = -50$, 0, and 80 ps respectively, where $t = 0$ is the time of peak emission. Circles in the frames indicate the positions where the light emission is plotted over time in panel d). Emission in d) is normalized by the peak of each curve.

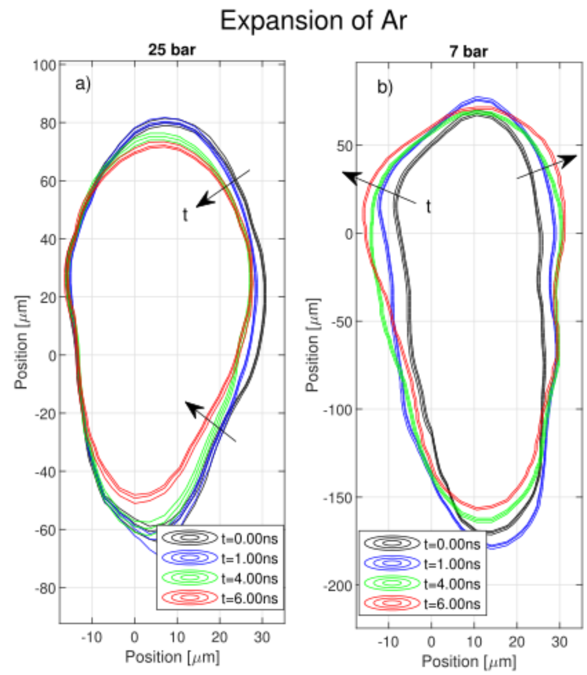


FIG. 4. Expansion of the breakdown plasma resolved by single shot CUP. The evolution of 25 bar argon a) and 7 bar argon b). Contours are drawn at $50 \pm 1\%$ of the maximum intensity at various times. The relative position of the contours indicates the expansion of the plasma. For plasmas formed at high density, the contours steepen and the plasma contracts, whereas at 7 bar the plasma expands.

1. Stefan-Boltzmann radiation from the argon plasma is small compared to the measured changes in the energy of the plasma.

2. Thermal conduction of energy to the surrounding ambient gas at 300 K is also small compared to the observed changes in energy.

It follows that the response is dominated by the internal thermohydrodynamics of a strongly coupled plasma. Furthermore, the plasma emission is not uniform and displays hot and cold spots. Rather than evening out, with hot spots getting cooler and cold spots warmer, the emission drops locally in a way that its profile maintains its shape during the cooling. This persistence of inhomogeneities within the plasma implies that the

3. Electron thermal conductivity inside the plasma is small compared to the timescale of the other transport processes.

4. Local processes dominate thermal equilibration between species. If $\tau_{e-th}$ is the time scale over which the temperature of electrons and ions equilibrate, we arrive at the same conclusion as Bataller [7] and interpret the rapid drop in emission at early times $t < \tau_{e-th} \lesssim 1$ ns for the higher pressures as arising from electron-ion equilibration [7, 17].

As the plasma cools and the electrons and ions equilibrate, some electrons will recombine and impart their ionization energy back to the electron gas causing recombination heating [18, 19]. But,

5. Signatures of recombination heating are not observed.

We speculate as to whether the effects of screening [20, 21] are much greater than is theoretically proposed [22, 23] for the parameter space realized in these experiments.

Certain regimes of SCP are predicted to have a tensile strength (negative pressure) [24]. Laser breakdown in lower pressure (density) gas immediately expands radially, consistent with ideal gas dynamics and positive pressure plasma. At higher pressure,

6. The plasmas display a lack of expansion, if not contraction, indicative of strong non-ideal interactions.

This suggests that these systems might be a route to investigating the hydrodynamic consequences of the tensile strength of SCPs.

## EXPERIMENT

The experiment (figure 5) uses a customized high-pressure chamber system built by Bataller [7, 25] which supplies high-purity noble gas with adjustable and accurate pressure. A Ti:Sapphire femtosecond laser (Coherent) centered at 800 nm passes through a 60 mm focal length lens to induce breakdown at the center of the gas chamber, where it arrives with an energy of 0.9 mJ. The laser beam propagates in the z-direction, is linearly polarized in the x-direction (see figure 5) and has a pulse duration tunable from 60 fs up to a few picoseconds. In this work, a temporally chirped pulse with 220 fs duration was optimized and chosen to generate a plasma with brightest emission (Supplemental Material figure S9). See the full characterization of the laser system in the Supplementary Material figures S5 and S6. The imaging system observes the plasma through the high-transmission quartz window on one side of the chamber. An infinity-corrected objective lens in conjunction with a tube lens forms an intermediate image, which is split into two copies by a beam splitter: one is recorded by a conventional CMOS camera (static view, s.v. for short) and the other is further processed to form the time-sheared dynamic view (d.v. for short). In this latter beam path, a digital micro-mirror device (DMD) (Texas Instruments) encodes the image by displaying a computer-generated 2D static pseudo-random binary pattern. Thanks to the modulation mechanism of tilting mirrors, two complimentary encoded images are reflected by the DMD in two separate directions. They are subsequently relayed and rerouted to a streak camera (Hamamatsu) side by side without overlap. Inside the streak camera, temporal shearing of the frames in the dynamic scene achieves from 1 million fps to 1 trillion fps. Signal synchronization between the laser and the streak camera is critical for the acquisition of any transient event. Eventually, one single fs-laser pulse generates one single spatially-encoded temporally-sheared raw image of one complete evolution of 2D plasma dynamics on the streak camera. Images are $250 \times 100$ pixels, and each pixel corresponds to a $3.5 \times 3.5$ $\mu$m plasma area. See Supplemental Material for experimental setup and components.

Emission spectra are measured through narrow-band filters centered at 410 nm, 515 nm, and 620 nm, and a broad-band filter covering the entire visible light (350-650 nm). The first three are extensively exploited to characterize the temperature and emissivity properties of blackbody radiation from plasma.

Important to the interpretation of the data is an auxiliary view (purple path) introduced here which is taken from another window and projected directly to an unused region on the streak camera. It is spatially-integrated and time-sheared and provides a key constraint on image reconstruction. In addition, a small portion of the original laser pulse is frequency doubled to a 400 nm second harmonic (SH) pulse by a nonlinear BBO crystal. This SH pulse flood illuminates the chamber through its back window, which is essential in imaging and temperature calibration and described in Supplemental Material §1.4.

Recovering and de-compressing the 3D spatiotemporal $(x, z, t)$ dynamics of the observed event from the two simultaneously acquired 2D images: one from the CMOS camera and one from the streak camera, is a typical ill-conditioned inverse problem due to multiplexing and high compression, the solution of which can only be reached by resorting to regularization [12]. Different from previous CUP modalities, the additional viewing channel introduced transforms the 3D data cube to a time sequence by 2D spatial integration. This added perspective towards the 3D domain presents a new angle of projection based on the theory of Radon transformation, potentially enforcing higher fidelity in reconstruction [12], and thus featuring a technological advancement from this work. The image forming pipeline, which is based on the assumption of data sparsity in the gradient domain, is discussed in the Supplemental Material §2.2.

## ANALYSIS

Temperature and emissivity $\epsilon$ are determined by fitting the emission of the brightest 7x7 pixel region of the plasma in the 3 listed windows. In general, the measured emission is $\bar{I}_\lambda = \epsilon_\lambda / [\exp(hc/\lambda kT) - 1]$ where the emissivity $\epsilon_\lambda$ is unity for a Planck blackbody but in general depends on wavelength. One restriction on $T$ is that $\epsilon_\lambda \leq 1$. In other words, there is a minimum $T$ at which one of the colors has $\epsilon = 1$ with the other wavelength having $\epsilon_\lambda < 1$. This value of $T$ is plotted in figure 6 as a lower bound. Motivated by the gray body approximation where $\epsilon_\lambda$ is independent of wavelength, we choose a higher temperature which minimizes the spread in $\epsilon_\lambda$. This temperature along with the various $\epsilon_\lambda$ is also plotted in figure 6.

A key parameter is the degree of ionization $x$. It determines the kinetic and ionization energy of a gas consisting of ions, free electrons, and neutral atoms:

$$E = \frac{3}{2} n_0 k(T_i + xT) + n_0 x \chi_0, \tag{1}$$

where $n_0$ is the number of atoms in the plasma, $T_i$ is the ion/neutral temperature, $\chi_0$ is the tabulated ionization energy, and we have allowed for the possibility that the electron temperature $T$ has not yet equilibrated with the ions/neutrals. To obtain $x$ from the spectral data requires a higher level of theory as it involves an analysis

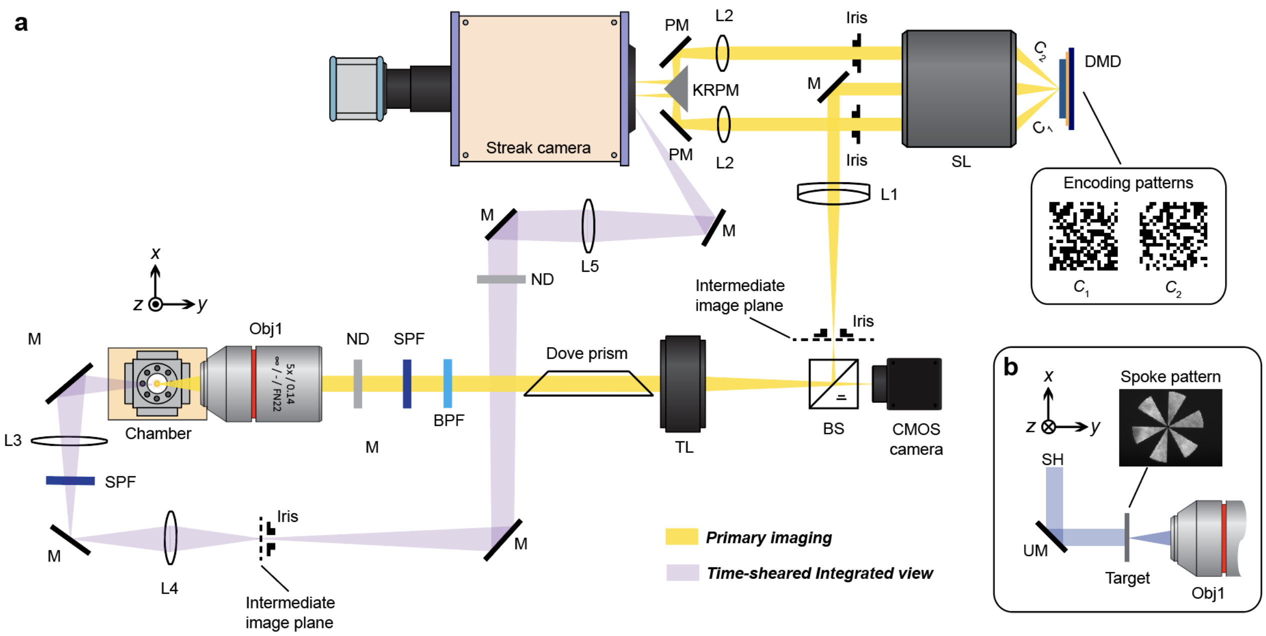


FIG. 5. Experimental CUP setup. a) Laser pulse propagating in the 'z' direction causes breakdown in the chamber. The yellow path provides spatially dependent emission to CMOS camera and Streak camera. Prior to time shearing by the streak camera the signal passes through a coded aperture provided by a DMD. The intermediate image plane, DMD and streak camera photocathode are conjugate planes. b) The second harmonic of the fs laser is used to illuminate a test target for the purpose of measuring the response of the coded aperture acquisition and CUP inversion to a delta function, which forms the basis for selection of the optimal mask (see Supplemental Material §1.11). Various optical elements such as band pass filters (BPF) and prisms (PM) are employed and listed with labels in Supplemental Material table S1.

of opacity and emissivity. We start with the Nekrasov formula [26] which connects the emissivity of a cylinder with diameter $D$ (estimated as the full-width-at-half-max emission) to the attenuation of light of wavelength $\lambda$ ($\kappa_\lambda$) in the plasma, $\epsilon_\lambda = 1 - \exp(-\kappa_\lambda D)$. In view of the featureless spectra observed here and in previous experiments [7, 27] we interpret the attenuation $\kappa$ in terms of inverse Bremsstrahlung which is free-free scattering. In this case a Drude model [28] relates $\kappa_\lambda$ to a frequency dependent electron collision time $\tau_\omega$ [7],

$$\kappa_\omega = \frac{k\gamma}{\omega\tau_\omega} = x^n A \ln(1 + \frac{B}{x^{n-3/2}}); \quad \gamma = \frac{\omega_p^2}{\omega^2} = \frac{4\pi n_0 x e^2}{m\omega^2} \tag{2}$$

$$A = \frac{2}{\sqrt{6\pi}}\frac{\omega}{c}\frac{\omega_{p,1}^3}{\omega^3}\Gamma_1^{3/2} f(\bar{\omega}); \quad B = \frac{0.7}{\sqrt{3} f(\bar{\omega})^{3/2}\Gamma_1^{3/2}} \tag{3}$$

$$f(\bar{\omega}) = \frac{1 - \exp(-\bar{\omega})}{\bar{\omega}}; \quad \bar{\omega} = \frac{\hbar\omega}{kT} \tag{4}$$

where $m$ and $\Gamma$ are the mass and plasma coupling constant of an electron. When $x > 1$: $\Gamma = xe^2/a_0kT$; the subscript "1" means evaluated at $x = 1$; for $x < 1$, $n = 2$ and for $x > 1$, $n = 3$. Emission and degree of ionization are determined by the electron temperature $T$. To obtain $x$ via opacity also requires the plasma diameter $D$ which is supplied by the radial lineout of intensity of emission as displayed for a single shot in Figure 7a. Values of $D$ and also $x$ are displayed in figure 6 as well.

## 1. BLACKBODY RADIATION IS SMALL

Blackbody radiation per unit surface area of the plasma is $\sigma T^4$ where $\sigma$ is the Stefan-Boltzmann constant. In the approximation that the plasma is a cylinder with diameter $D$ and length $L$, the loss in energy due to radiation is $dE/dt|_r = -\pi D L \sigma T^4$. The plasma energy will have contributions from the kinetic energy of ions, electrons and neutrals, as well as the potential energy of ionization. We shall first consider 25 bar argon at times after 2 ns so as to exclude the initial rapid drop in emission which is ascribed to electron-ion thermal equilibration. Then the kinetic energy per unit volume is $(3/2)n_0(1 + x)kT$ where $n_0$ is the density of the ambient, 300 K, 25 bar gas. Cooling due to blackbody radiation is $dT/dt|_r = -8\sigma T^4/[3kDn_0(1 + x)]$. Using $n_0 = 6.2 \times 10^{20}$ cm$^{-3}$ and $D = 43$ $\mu$m yields cooling at a rate of 110 K ns$^{-1}$ for parameters of $x = .9$ and $T = 27$ kK that apply at 2 ns. A fit to the temperature in figure 6 gives,

$$T(t) = T(0) + A\left[e^{-t/\tau} - 1\right] - Bt, \tag{5}$$

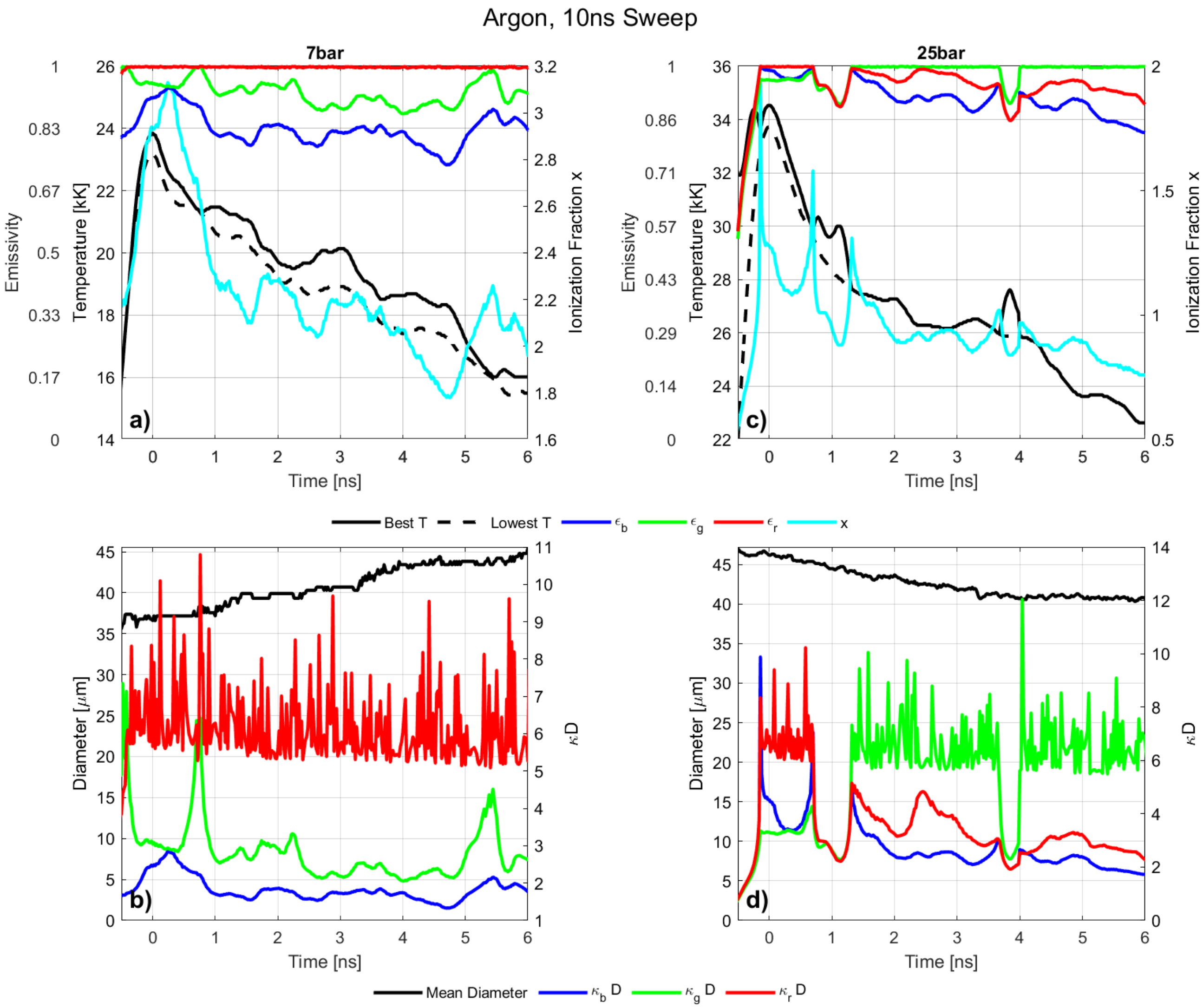


FIG. 6. Temperature, emissivity, ionization, mean diameter $D$, and optical thickness ($\kappa D$) for breakdown in 7 bar (a,b) and 25 bar (c,d) argon for 10 ns sweep. Temperature is based upon emission from the central region of the plasma. Emissivities $\epsilon_r$, $\epsilon_g$, $\epsilon_b$ are displayed for the best fit temperature. The dashed line is the minimum temperature at which the separate emissivities are all $\leq 1$. The degree of ionization is determined from the lowest emissivity which is generally the blue curve. The singularity in transport properties which accompanies emissivities close to unity leads to the spikes that can be seen in the curves for $\kappa D$, which is the photon mean free path.

where $T(0) = 3.05$ eV, $A = 0.5$ eV, $B = 0.1$ eV ns$^{-1}$, and $\tau = 0.33$ ns so that the observed cooling rate is $B \simeq 1100$ K ns$^{-1}$. The 10× higher cooling rate observed implies that radiative cooling is small and that other processes are more important. Inclusion of the ionization potential $\chi_0 = 15.76$ eV would increase the heat capacity and further decrease the magnitude of the calculated cooling should $x$ be a decreasing function of $T$.

## 2. CONDUCTION TO AMBIENT GAS

Thermal conduction to the exterior ambient gas is limited by the thermal conductivity of ambient argon. The maximum heat flow is realized under the assumption that the plasma temperature is constant with a step function initially at the boundary. In this case the heat flow leads to a cooling given by [29],

$$\frac{\Delta T}{T} = -\frac{5}{6(x+1)}\sqrt{\frac{\chi_T t}{\pi D^2}}, \tag{6}$$

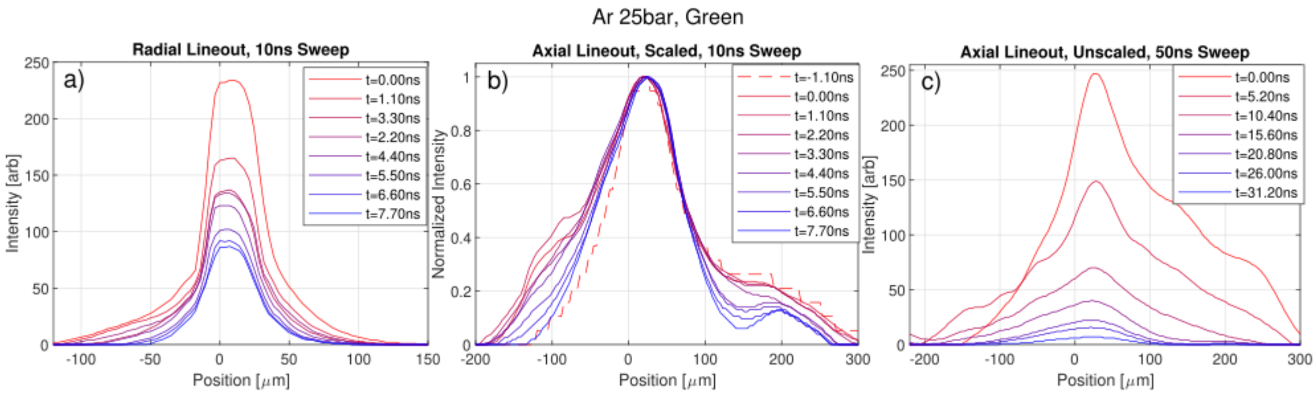


FIG. 7. Emission of a single shot along a lineout crossing the brightest pixels for argon 25 bar acquired through the green filter. a) is a radial lineout for a 10 ns sweep, b) is an axial lineout for a 10 ns sweep that is scaled to the peak value, and c) is an axial lineout for a 50 ns sweep. The rapid drop in emission in (a) during the first nanosecond is due to local equilibration of electrons, ions, and neutrals. Notice that the central peak in (b) maintains its curvature for at least 10 ns, which indicates the low level of thermal conduction within the plasma.

where $\chi_T \approx 6 \times 10^{-4}$ $\mu$m$^2$ ns$^{-1}$ is the thermal diffusivity of the ambient argon which surrounds the plasma and limits the heat flow to the environment. We are using the same plasma model as for the discussion of radiative cooling. For $t = 4$ ns and $x = 0.6$ this yields $\Delta T/T = 3 \times 10^{-4}$ which again is small compared to the observed response of the plasma.

## TRANSPORT WITHIN THE PLASMA

Turning now to a discussion of transport time scales within the plasma we consider various cases for the collision cross-section $\sigma$:

a) $\sigma_{e,dilute} \approx Ar_T^2 \ln \Lambda$ is the cross-section for an electron-electron or electron-ion collision in a low $\Gamma$ or hot plasma [30], where $r_T^2 = \left(\frac{e^2}{kT}\right)^2$ is the impact parameter within which the collision is thermodynamically meaningful and $\Lambda = \delta_D/r_T$. Following Stanton and Murillo and Daligault [17, 31], $A = 8\sqrt{2\pi}/3$. As an example, at 15,000 K and $x = 1$ we have $r_T \sim 1.1$ nm. $\delta_D = \sqrt{kT/8\pi x n_0 e^2}$ is the Debye screening length, which is much greater than $r_T$ in the dilute limit.

b) $\sigma_{e,D} \approx \frac{32\sqrt{2\pi}}{3}\delta_D^2 = \frac{32\sqrt{2\pi}}{3}\frac{r_T^2}{6\Gamma^3}$ is the strong screening limit which applies when $\Gamma$ is large. The expression will be used in the context of electron-electron transport processes where $n_e = x n_0$. A numerical factor of 4 has been introduced so as to match the unifying transport formula of Stanton and Murillo [31].

c) $\sigma_{e-int} = \frac{8\sqrt{2\pi}}{3} r_T^2 \ln\left(1 + \frac{b_0}{\sqrt{x}}\right)$ which applies to the intermediate values of $1 \lesssim \Gamma < 3$ and $b$ is evaluated at $\bar{\omega} = 0$. This equation will be used for electron-electron and electron-ion collisions [17].

d) $\sigma_{e-0} = 10^{-16}$ cm$^2$ at 1 eV [32] describes collisions of electrons with neutral argon atoms, with a collision time $\tau_{e-0} = 1/v_{th}(1-x)n_0\sigma_{e-0}$ where $v_{th} = \sqrt{kT/m}$ and $m$ is the electron mass. This expression applies when ionization is low.

## 3. THERMAL CONDUCTION WITHIN PLASMA

Aside from conduction to the ambient gas, one can ask how heat moves to different locations within the plasma. In particular, how does thermal conduction within the plasma affect heat flow along the direction $z$ of the laser pulse where maximum inhomogeneity is expected. Figure 7 shows lineouts through the region of peak emission at $r = 0$ and $z = 0$. Panel a) displays a $\sim 3\times$ drop in emmision during the first 10 ns. Despite this, as shown in panel b), the curvature of the scaled emission profile barely changes with time, so that even within the plasma thermal transport has a small effect on the emission profile on the resolved timescales. Although electrons are efficient transporters of heat [33] this observation is consistent with theory.

A general expression for the electron thermal diffusivity is $\chi_{T-e} = \frac{2\nu_{th}}{3\sqrt{\pi}\sigma n}$ where $n$ is the density of scattering sites. The strongest electron thermal conduction is realized when the cross-section is given by case b). As an estimate, take $x \sim 1$ and $T = 30$ kK, $n_0 = 6.2 \times 10^{20}$ cm$^{-3}$, $\nu_{th} = 6.9 \times 10^7$ cm s$^{-1}$, $r_T = 5.3 \times 10^{-8}$ cm so that $\Gamma \gtrsim 0.7$, $\sigma = 2.5 \times 10^{-14}$ cm$^2$ to get $\chi_{T-e} = 0.17$ $\mu$m$^2$ ns$^{-1}$. The mean free path is $\ell \sim 6.5 \times 10^{-8}$ cm. Consider the diffusive spread in time of a Gaussian fit to the data. The width $w(t)$ at 80% of the maximum should expand as $w^2(t) = w^2(0) + 3.6\chi_{T-e}t$. Using $w(0) \approx 60$ $\mu$m yields

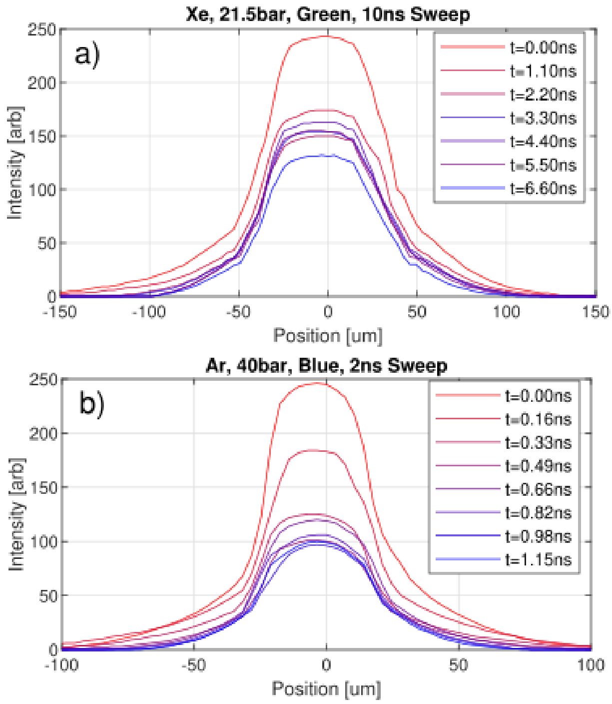


FIG. 8. Emission along a radial lineout through the brightest central pixels for a) xenon at 21.5 bar and b) argon 40 bar. The rapid drop in emission during the first nanosecond is due to local equilibration of electrons with ions and neutrals.

an increase in width less than 1 $\mu$m after 10 ns. This is consistent with experimental measurements which show that there is a very slight change in width, if not a contraction, during the first 7 ns. We return to this issue in §7 below.

## 4. ELECTRON-ION EQUIPARTITION OF ENERGY

The laser directly excites the electrons on a femtosecond timescale and so creates a situation where the ions have an initial temperature $T_i < T$. The subsequent equilibration leads to a rapid drop in emission, which we interpret as a proxy for temperature, during the first nanosecond. As we have argued, there is insufficient heat flow to the ambient gas, and thermal conduction within the plasma is low as well. This leads one to interpret this drop as being due to electron-ion equipartition of energy. After equilibration the drop in emission is slower. This behavior is described with cross-section c) where collisions are partially screened [17, 28], the parameter space for these experiments. When applied to electron-ion collisions the time scale for thermal equipartition will be increased by the ion/electron mass ratio to $\tau_{e-th} = (M/m)\tau_{e-ion}$ where $\tau_{e-ion} = 1/n_i\sigma_{e-int}v_{th}$, $M$ is the ion mass and $n_i = xn_0$ is the density of ions (taking here $x < 1$). This early time equilibration is qualitatively apparent in figure 8 and the best fit to the temperature (equation (5)) yields a time scale of about 300 ps which should be compared with the 200 ps estimated with cross section c).

Ions can also be heated above their initial ambient temperature of $\sim$ 300 K by a process which is much faster than the collisional equilibration discussed above. In our experiment, the laser pulse randomly rips electrons off the atoms on the femtosecond time scale. In this initial plasma state the ions have a 2 particle correlation $g(r) = 1$. This function describes the probability of having two ions a distance $r$ apart. However, $g(r) = 1$, does not match the values in thermodynamic equilibrium under the given plasma coupling parameter, $\Gamma_i = e^2/akT_i$ for $x < 1$ or, $\Gamma_i = Z^2e^2/a_0kT_i$ for $Z \geq 1$ where $Z$ is the degree of nuclear ionization. For the case at hand of 25 bar argon, $x = 0.73$ at $t = 0$, so $\Gamma_i \sim 65$. This high value of $\Gamma_i$ means that the ions instantaneously constitute a very strongly coupled plasma which is out of the equilibrium determined by the non-local Coulomb forces. The approach to equilibrium results in a "disorder induced heating" (DIH) [5, 34] which occurs on a picosecond time scale which is much shorter than $\tau_{e-th}$. DIH would lessen the magnitude of the drop in emission that accompanies electron-ion equilibration.

According to Figure 7b emission decreases more rapidly for $z \sim 200$ $\mu$m with a decay rate that is approximately 2-3 times more rapid than at the peak. We interpret this rapid decay as being due to lower ionization with a corresponding increased contribution from the neutral atoms. In this case the effective cross-section is described by d). This observation is made possible by the slow axial flow of heat.

## 5. SCREENING AND LEVEL OF IONIZATION

Based upon emissivity which we interpret as arising from the scattering of light by free charges [7, 20, 33, 35–37], the deduced degree of ionization $x$ can be much larger than what follows from Saha's equation [20, 21, 35] even when leading order screening effects [22, 23] are included. Consider figure 9, which show the breakdown emission response of argon gas at 7 bar taken under the same physical conditions as figure 6, but for longer sweep or period of data acquisition. Since the ionization energy for argon is $\chi_0 = 15.76$ eV and the temperature is 2 eV or less, one would expect that the plasma is single ionized at most. This supposition is confirmed by Saha's equation of ionization,

$$\frac{x^2}{1-x} = \frac{2g}{n_0\lambda_{deB}^3}e^{-\chi/kT}, \tag{7}$$

where the thermal de Broglie wavelength is $\lambda_{deB} = h/\sqrt{2\pi mkT}$, $n_0 = 1.7 \times 10^{20}$ cm$^{-3}$, $a_0 = 1.14$ nm and

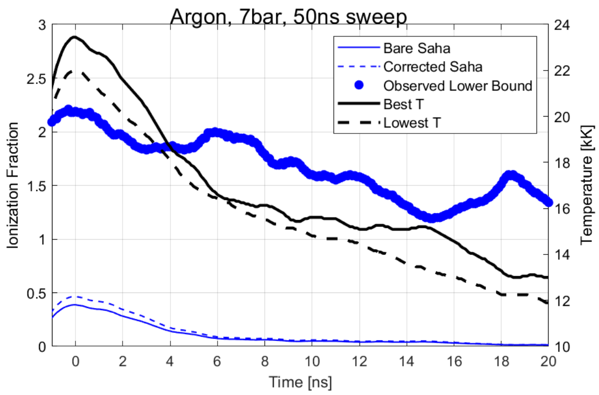


FIG. 9. Temperature and ionization level of 7 bar argon as compared to Saha's equation (7). The thick blue line is the level of ionization as determined from emissivity and transport theory under the assumption that the photon mean free path is due to inverse Bremsstrahlung. The thin solid blue curve is the level of ionization calculated with Saha's equation and the dashed blue curve is calculated with the first order screening effects included. Both calculations are well below the lower bound, which suggests a larger screening effect or some other large source of continuum lowering.

for argon the degree of degeneracy $g = 6$. According to Saha's equation, $x = 0.4$ at 2 eV ($t = 0$) and $x = .07$ at 1.4 eV which is the temperature at 7.0 ns. The value of ionization deduced from opacity to blue light is much larger, being $x = 2.2$ ($t = 0$) and 2.0 ($t = 7$ ns). As blue emissivity is lowest, we have used that value for this calculation. The value of $x$ calculated from red and green emissivities is regarded as a lower bound as these values are close to one.

Saha's equation applies to an isolated atom and this version applies when only neutrals and singly ionized atoms are present so that doubly ionized atoms are inconsequential. In a dense gas the ionization potential is lowered by screening which leads to an effective ionization potential [22, 23],

$$\chi = \chi_0 - \frac{kT}{2}\left((1+\Lambda)^{2/3} - 1\right), \tag{8}$$

where $\Lambda = \left(3x^{1/3}\Gamma_1\right)^{3/2}$ (for $x < 1$). Calculations of the level of ionization based on the bare (7) and supplemented Saha equations (8) are plotted in solid and dashed blue curves in figure 9. Both lie well below the minimum level of ionization observed and cannot account for the observed opacity based upon the attenuation of light being due to inverse Bremsstrahlung. Either opacity is due to different dynamics, or some new processes [35] is contributing to a further lowering of the ionization energy.

## 6. LACK OF RECOMBINATION HEATING

Figures 6 and 9 show the cooling of a 7 bar argon plasma at a rate of 1100 K ns$^{-1}$ between 0 and 5 ns. Let us compare this to the average cooling due to blackbody radiation at the intermediate temperature of 20,000 K at 2 ns where $D = 40$ $\mu$m. This would lead to a drop in temperature of 90 K ns$^{-1}$ when the heat capacity is that of an ideal gas. This difference is accentuated when the possibility of recombination heating is evaluated. From equation (1) a change in the degree of ionization $\Delta x = x(0) - x_f$ also would not only inhibit cooling but lead to heating as implied by energy balance:

$$\frac{3}{2}(1+x_f)kT_f = \frac{3}{2}(1+x(0))kT_0 + \Delta x\chi_0 + \frac{1}{N_0}\int_0^f \frac{dE}{dt}|_r dt, \tag{9}$$

where subscripts $f$ and 0 refer to final and initial states. The last term is the radiated energy and for the purpose of developing this argument we are using the bare ionization potential. Figure 9 shows that the degree of ionization, as determined by opacity to blue light, decreases by $\Delta x = 0.6$ during the interval $0 < t < 10$ ns. Using $T(0\text{ ns}) = 23,600$ K, and $\Delta x = 0.6$ leads to $T_f = 2.4T(0)$ when $\chi_0 = 15.8$ eV. This is a very large heating and there is no such signature in the data. We propose that a larger-than-expected reduction in the ionization potential could account for both the high ionization and weak recombination heating.

There are other sources of cooling. An expanding plasma does work against the exterior gas. In view of the large pressure and temperature difference between the inside and outside of the plasma channel, this effect is small. The hydrodynamic kinetic energy developed in an expanding plasma has also not been included in the energy balance. Increases in this quantity during an expansion can lead to adiabatic cooling [38] which is discussed in the next section. When the region of emission expands, we propose a means whereby screening can lead to cooling. Emission from newly formed plasma requires local ionization which comes at the expense of recombination further inside the plasma, where the ionization energy is lower due to screening.

## 7. LACK OF EXPANSION

A possible source of cooling is the adiabatic expansion of a gas. Our experiment provides single-shot, time-resolved measurements of the plasma's two-dimensional macroscopic motion, enabling us to observe its changing shape over time as for instance indicated in figure 4. To what extent does this data give further insight into cooling and hydrodynamic motions?

In a simple model, laser breakdown creates an initial condition with a spatial discontinuity in pressure an temperature at constant mass density. According to the Euler equations, such a contact discontinuity generates an outgoing shock wave as well as an expanding entropy (temperature) wave. In an ideal gas in one dimension, such an entropy discontinuity initially propagates outward at the speed $u/2\gamma$, where $u(T, x)$ is the speed of sound on the hot side, and $\gamma$ is the adiabatic index. The supplemental material shows this effect for the case where $T/T_0 = 100$ and $\gamma = 5/3$ (see figure S13).

The initial state of the brakdown plasma also includes a discontinuity in ionization. So the ideal gas kinetic pressure is $p = n_0(kT_i + xkT)$ is higher than used in the above model. For 25 bar argon after electrons and ions have equilibrated, $T \sim 28$ kK and $x \sim 1$ leading to $p \sim 4800$ bar. Using the adiabatic coefficient $\gamma = 5/3$, a characteristic sound velocity is $u \sim \sqrt{\gamma p/n_0 M_{Ar}} \sim 4.4\ \mu\mathrm{m\ ns}^{-1}$. For 7 bar argon the characteristic temperature at 2 ns is 19 kK with $x \sim 2$ yielding $u \sim 4.4\ \mu\mathrm{m\ ns}^{-1}$.

Figure 4 compares plots of contours of constant emissions for 7 and 25 bar argon. The expansion of the radius $D/2$ of the waist of emission from the breakdown of 7 bar argon gas takes place at a speed of $0.7\ \mu\mathrm{m\ ns}^{-1}$ (see figure 6) which is about a factor of 2 less than the $u/2\gamma$ expansion of a neutral ideal gas with the same sound velocity. Figure 4a shows the contours of constant emission for 25 bar argon steepen, if anything, and do not propagate outward. This applies in all directions. For 7 bar the plasma does not expand in the $z$-direction which on average is 7 times longer than the waist is wide.

Regarding the lack of expansion at 25 bar, we note that one component plasmas can demonstrate a tensile strength [24]. Whether these experiments are observing evidence for such effects will require more detailed modeling; in which we note that the ionization plotted in figure 6 should be regarded as a lower bound in view of the small deviation of emissivity from one. Our time-resolved measurements suggest that we may be entering the regime where ion coupling and correlation effects play a significat role, which would provide a testing ground for various hydrodynamic [14, 16] or kinetic [39] models that incorporate these effects.

Self-similar solutions for the expansion of a plasma into a vacuum have been derived from hydrodynamics [38, 40] or kinetic [41] equations for an ideal plasma. Although our initial plasma is not surrounded by vacuum, the ambient 25 bar is small compared to the plasma pressure. For Mora [38], both the atomic density and the electron density are Gaussian. Due to the femtosecond nature of the breakdown which we study, the atomic density is best regarded as flat. However, a self-similar solution is also possible where the electron density is Gaussian and quasi-neutrality is maintained by a constant mass density which has a compensating space and time dependent ionization fraction. In this model, the electrons expand at a velocity determined by the electron and not the ion mass. Such a rapid expansion is not seen.

In general, the volume of the plasma contracts rather than expands. For an ideal gas, this contributes to heating, rather than cooling. Typical hydrodynamic equations cannot explain the macroscopic motion.

## CONCLUSION

Here we point to the use of Compressed Ultrafast Photography in advancing the physics of a strongly coupled plasma for which the theory is incomplete. In this direction CUP has been used to resolve the time dependent 2-D behavior of a breakdown plasma formed from gases that are sufficiently dense that the plasma emissivity is close to unity. The key quantity measured is the plasma emission. Many physical processes contribute to the emission [42]. In order to facilitate an application of the CUP data we have interpreted emission and absorption according to the free-free theory of plasma radiation and absorption (Bremsstrahlung and inverse Bremsstrahlung). Plasma breakdown creates an ionized medium where the mean free path of light is due to its interaction with free electrons. Although this is in many ways the simplest model, we propose that it is in fact appropriate because detailed spectral measurements [27] are smooth and lack features. Its application points to a degree of ionization that can only be explained with a reduction in the effective ionization potential that exceeds existing theories. Temperature is determined by a graybody fit and we find that an argon plasma formed at 25 bar is an isolated system for about 10 ns during which time thermal conduction to the environment and radiative cooling are small. The two dimensional (radial and axial) response during electron ion equilibration is resolved. Free expansion is observed at low but not high gas density. As time progresses ionization decreases yet we do not discern signs of recombination heating. We envision extension of this method to study 2D hydrodynamics especially during the time scales where $\Gamma > 1$. For 7 bar argon at 5.5 ns, $T = 16$ kK, and $x = 2$, yielding an ion coupling coefficient $\Gamma > 3$. For these parameters the Debye screening length $\delta_D = 0.34$ nm and $\delta_D/a_0 = 0.3$ so that the ratio of Thomas Fermi to Debye screening lengths is $\delta_{TF}/\delta_D \simeq 0.4$. Figure 1 shows just a few frames from a movie with over 250 frames of a single event with a duration of 500 ps. These systems are then test-beds for theories of equations of state and strongly coupled hydrodynamics, and sources for the unmasking of emergent properties of these systems. It appears reasonable to apply this technique to obtain a single shot movie of processes that have been collected frame by frame using the pump/probe technique [43] and to extend its use to x-ray-CUP of dynamical processes in dense systems [44].

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# Electron Recombination Time Measurement in High Density Sulfur Plasma

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## Introduction

All plasmas exist as a mixture of electrons, ions, and neutral particles, and understanding how the proportions of these constituents change in time is of vital importance when describing their dynamics and chemistry. In thermal equilibrium, these quantities are easily derived from macroscopic variables via the plasma's equation of state: the ratio of the electron density to neutral density is determined by Saha's Equation. However, when a plasma is off of its equilibrium, such as when temperature or pressure changes rapidly, it takes a finite amount of time for the plasma to reach a new equilibrium. The characteristic time in which a plasma reaches its new thermal equilibrium is its recombination time. This time scale is a crucial limiting factor on the rate at which some chemical processes can occur in the plasma. Also, the possibility of creating a self-oscillator hinges on the recombination time being substantially smaller than the period of an oscillation [1]. To our knowledge, this paper represents the first effort to quantify this recombination time for a quiescent plasma of atmospheric density. The theory of recombination is explored by Zeldovich and Raizer [2], whose formulation estimates the electron recombination rate by determining the rate of three-body (two electrons, one ion) collisions. Our experiment convincingly verifies their theory. Most experiments that have measured plasma recombination times in the past have done so for highly rarefied plasma, where the lack of collisions results in recombination times on the order of hundreds of microseconds to milliseconds [3]-[6]. Similar measurements for dense plasmas have found recombination times on the scale from microseconds to nanoseconds [7]-[9], but these experiments were performed on laser or pulse induced plasmas. Our experiment uses a similar set up to Johnston et al. [10], who explored the use of power interruption measurements in Sulfur plasma lamps to make off-equilibrium relaxation time measurements in a uniform volume of high-density plasma. We employ the same method to not only derive a recombination time of Sulfur plasma, but to explore new methods of measuring intensive thermodynamic plasma properties in real time.

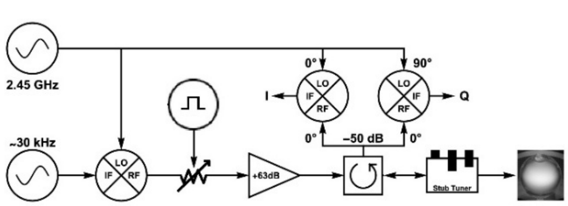


Fig. 1. A block diagram of the plasma heating and measuring system. Mixer ports are LO: local oscillator; IF: intermediate frequency; and RF: radio frequency.

Finding the recombination time requires one to measure both the actual charge density of a plasma as well as its equilibrium value in real time, then to compare their different rates of change by the Landau method of analyzing second viscosity [11]. Our experiment accomplishes this with real time measurements of the plasma's complex reflection coefficient, whose real and imaginary components can be used to derive the charge density and temperature [12]. The temperature, in turn, yields the ideal charge density. Furthermore, we developed an acoustic method for real-time temperature measurement of the plasma, allowing us to compare the efficacy of the reflection coefficient method. We were able to verify our method, opening the door for precision real-time temperature measurements for plasmas with recombination times much faster than acoustic periods. Our ultimate findings show that the recombination time of a dense plasma is well predicted by three-body recombination as outlined by Zeldovich [2]

## Experiment

Fig 1 shows the block diagram of our plasma system. An Empower 2180 programmable amplifier, with a top output power of 2kW, generates the microwave signal that can simultaneously heat the plasma and drive it at resonant acoustic frequencies. The signal is carried by a WR340 waveguide to a cylindrical cavity made of metallic mesh that confines the microwaves but allows for visualization. Within, a spherical quartz bulb containing 35 mg of sulfur is held in the center and spun at ~70 Hz. The signal is impedance matched to the waveguide by a three-stub tuner. The signal is comprised of a 2.45 GHz carrier wave that can be modulated at acoustic frequencies to drive resonant oscillations. Running at full power, our system generates a quiescent Sulfur plasma with a neutral density of $\sim 10^{19}\ cc^{-3}$ at a temperature of $\sim 4{,}000\ K$.

Our system collects real-time data from two primary sources: a photodiode and the reflected signal from the amplifier. The photodiode records the total brightness of the bulb and the reflected power is measured in quadrature as the in-phase and out-of-phase reflected signals relative to the input signal. An additional phase shift is incurred due to distance traveled in the circuit and must be accounted for in our analysis.

### Acoustic Temperature Measurements

We must be able to measure temperature as a function of time in order to determine the equilibrium charge density as a function of time. We do this acoustically by observing the change in the acoustic resonance frequency of the bulb as a function of time, while the plasma undergoes power

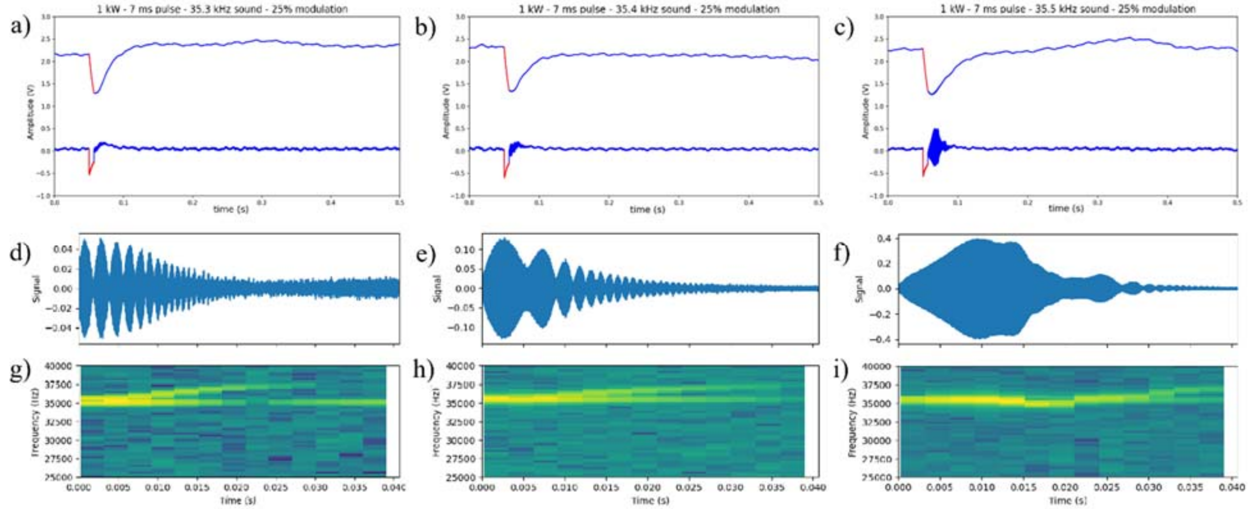


Fig. 7. The results of a series of power interruption measurements of the plasma to determine its acoustic resonance after cooling. All interruptions from the runs shown were of the same duration (7 ms) but with different frequency of acoustic modulation. Panels a.-c. show the photodiode data measured for an interruption of a given acoustic modulation frequency. The top line represents the low-pass-filtered (at 10 kHz) "DC component" of the photodiode while the bottom line is the high-pass-filtered (10 kHz) "AC component". The red highlight represents the period when power was interrupted. Panels d.-f. depict a zoomed in view of the high frequency resonant oscillations from the AC component of the graph above it, and g.-f. are spectrum analyses of these oscillations respectively.

interruption. Assuming that the quiescent plasma has a uniform temperature and density profile, the speed of sound ($c_s = \sqrt{\gamma k_b T/m_e}$) is purely a function of temperature. For a spherical geometry, the frequency for the lowest spherical mode [1] is $\omega_{02} = 4.49 c_s/R$.

Photodiode measurements were taken of the plasma undergoing power interruption. All runs begin with a quiescent plasma receiving the full 2kW output of the amplifier and impedance matched via the three-stub tuner so that no reflected power is observed ($\Gamma \sim 0$). After measurements begin, the power is attenuated to 10% output. The power drop causes the plasma to begin cooling, primarily through radiation, but because the power is non-zero, there will still be a small outgoing signal that can reflect and return I/Q measurements for later use. After the pre-programmed interruption time passes, full power is restored, and the plasma is allowed to return to quiescence. Throughout all this process, the outgoing signal is modulated

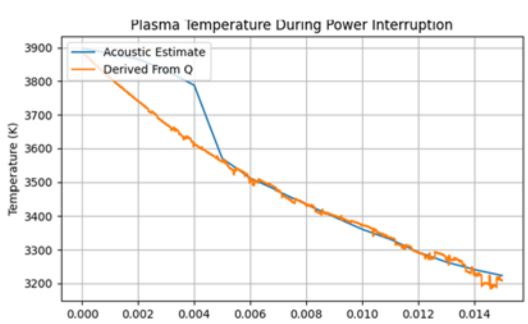


Fig. 3. The temperature of the plasma after experiencing a power interruption of a specified length. The blue line represents temperature estimated acoustically while the orange line is derived from our Q measurements.

at an acoustic frequency below the acoustic resonance frequency of the quiescent plasma.

Figure 2. shows a series of these measurements taken for a single duration of interruption and with different acoustic drive frequencies. What we observe is that photodiode output falls and rises with the temperature of the plasma, but for certain drive frequencies we see high frequency oscillations characteristic of resonant oscillations from a driven oscillator with a fixed drive frequency and a resonant frequency that is increasing with time (see Appendix). As the plasma heats up after the interruption, the resonant frequency increases from its minimum value, passing through the drive frequency on its way back to its quiescent value, generating visible resonant oscillations. For a set duration of interruption, we find there is a minimum drive frequency below which we no longer see resonance. This frequency is therefore an upper bound on the minimum resonance frequency attained by the plasma and can therefore be used to place an upper bound on the lowest temperature reached in that time span. When repeated over multiple timespans, one can thus find an upper bound on plasma temperature as a function of time as in Figure 3. Furthermore, if we run a spectrum analysis on these resonant oscillations as shown inf Fig. 2.g-i., we can see a delineation between the constant acoustic drive line and the increasing plasma resonance line that seem to perfectly meet at the origin when the drive frequency is tuned to this minimum threshold frequency (35.4 kHz in the figure shown). This further indicates that upper bound on temperature found in Figure 3. is not just an upper bound but rather a direct measurement of the temperature.

It should be noted, however, that these measurements are less reliable at predicting short interruption temperatures (~5ms) as there is insufficient time to observe a resonance.

## Measuring Instantaneous Plasma Properties from the Reflection Coefficient

The complex reflection coefficient can be used to measure instantaneous plasma properties [12]. This coefficient, $\Gamma$, is often expressed in quadrature as

$$\Gamma = E_{ref}/E_{inc} \propto I + iQ \qquad (1)$$

where $E_{ref}$ is the reflected wave; $E_{inc}$ is the incident wave, and I and Q the in-phase and out-of-phase reflected signal respectively.

It can be shown that the reflection coefficient of waves incident on a lossy cavity vanishes for oscillations near the resonance frequency with damping matching the cavity coupling. Under these circumstances, small deviations in the drive frequency or the damping yield

$$\Delta\Gamma \approx \frac{\Delta\lambda}{\beta} - i\frac{\Delta\omega_r}{\beta} \propto Im[\Delta\omega^*] + i\,Re[\Delta\omega^*] \qquad (2)$$

where $\omega_r$ is the resonance frequency; $\lambda$ is the damping within the cavity; $\beta$ is the coupling of the cavity; and $\omega^* = \omega_r - i\lambda$ is defined as the complex resonance frequency.

The plasma's complex permittivity, $\epsilon$, is [13][14]

$$\begin{aligned}\frac{\epsilon}{\epsilon_0} &= 1 - \frac{\omega_p^2}{\nu_c^2+\omega^2} - i\frac{\nu_c}{\omega}\frac{\omega_p^2}{\nu_c^2+\omega^2} \\ &\simeq 1 - \frac{\sigma}{\epsilon_0\omega}\frac{\omega}{\nu_c} - i\frac{\sigma}{\epsilon_0\omega}\end{aligned} \qquad (3)$$

for drive frequency $\omega$; plasma frequency $\omega_p = n_e e^2/(m_e\epsilon_0)$; electron density, charge, and mass, $n_e$, $e$, and $m_e$ respectively; collision frequency $\nu_c = n_e^2\sigma_{ie}\sqrt{8k_BT/\pi m_e} \sim 500\,GHz$; temperature T, collision cross section $\sigma_{ie}$ [15][16] and conductivity $\sigma = \omega_p^2\epsilon_0/\nu_c$. The second line follows if $\nu_c \gg \omega$, which is well satisfied in our experiment. Microwave cavity perturbation theory of our cavity also yields

$$\frac{\Delta\epsilon}{\epsilon_0} = -\frac{\Delta\omega^*}{\omega_0}\left(\frac{1-\eta}{\eta} + \frac{\epsilon_q}{\epsilon_0}\right) \approx -\frac{\Delta\omega^*}{\eta\omega_0} \qquad (4)$$

where $\eta$ is the effective volume fraction of the quiescent plasma ($\eta = 0.003 - 0.009$ and $\sigma_0 = 0.065 - 0.19\,\Omega^{-1}m^{-1}$ in our set up according to simulation) and $\epsilon_q$ is the average quiescent permittivity. The second equality holds when $\frac{\epsilon_q}{\epsilon_0} \approx 1 - i$ is small compared to $\frac{1-\eta}{\eta} \approx 200$.

Combining (3) and (4) with (2) while eliminating small terms ($\sigma\Delta(1/\nu_c)$ is orders of magnitude larger than $(1/\nu_c)\Delta\sigma$) yields

$$\frac{\Delta\sigma}{\sigma_0} \approx \frac{\omega_0\epsilon_0}{\eta\sigma_0} Im\left[\frac{\Delta\omega^*}{\omega_0}\right] \propto I \qquad (5)$$

$$\Delta\frac{\omega}{\nu_c} \approx \frac{\omega_0\epsilon_0}{\eta\sigma_0} Re\left[\frac{\Delta\omega^*}{\omega_0}\right] \propto Q \qquad (6)$$

This means that, holding the drive frequency constant and the cavity near resonance, measuring Q allows us to measure the collision frequency and measuring I allows us to measure the conductivity (which is a function of both charge density and collision frequency). Furthermore, the collision frequency is a function of temperature, allowing us to derive temperature measurements from Q.

The above describes the components of the wave reflected in and out of phase at the plasma cavity, but the raw data measured by our equipment measures the phase relative to the outgoing signal from the function generator and thus includes an additional phase shift due to the difference in distances traversed by the function generator signal and the reflected plasma signal. To calibrate our equipment to remove this phase shift, we first power the plasma with a signal tuned to its complex resonance but with a small frequency modulation at 50 kHz on top of it. Since small modulations of frequency only affect the real component of the complex resonance, this 50 kHz oscillation should only be visible in the Q signal. Thus we can correct the I and Q signals by adding a phase shift until the 50 kHz is measured entirely in Q. Figure 4. shows a fourier transform of the calibration signal before and after the rotation.

Figure 5.a shows the calibrated I and Q measurements of a power interruption experiment. Here, because the power is non-zero, we can measure the reflection of the small outgoing signal to evaluate $\Delta\Gamma$. The changing temperature

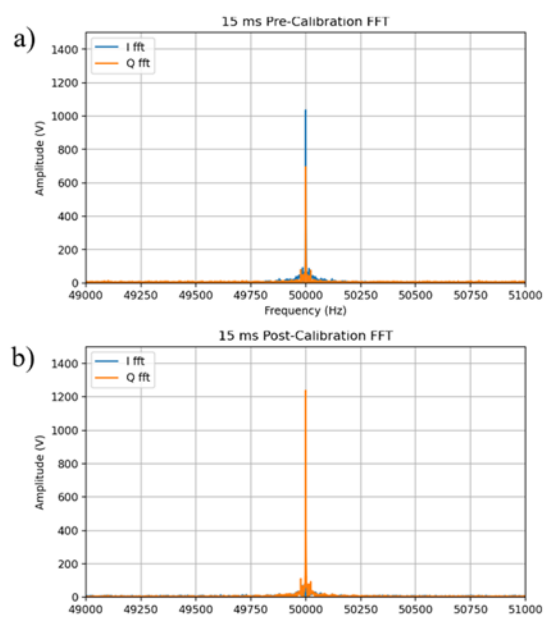


Fig. 4. FFTs of the reflected 50 kHz signal from a quiescent plasma measured in phase (I) and out of phase (Q) relative the drive. In a) we see the signal received by our equipment while b) shows the signal after a rotation transformation to minimize the signal in I.

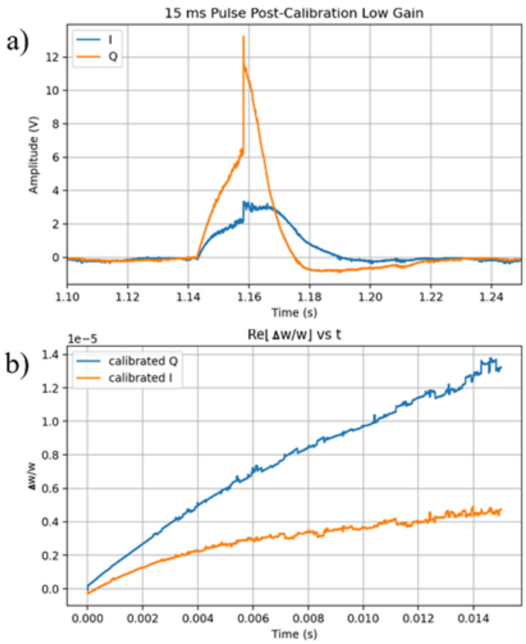


Fig. 5. The calibrated I and Q measurements of the plasma undergoing a 15 ms power interruption. a) depicts the measured signal strength of the whole event, with the interruption beginning at 1.143 s and ending at 1.158 s while b) shows the I and Q in units of fractional change in frequency during the power interruption.

will drive changes in the collision frequency and charge density detectable in I and Q. For these measurements, the signal was not modulated by an acoustic frequency. The end point of the power interruption is clearly visible in Figure 5.a as the discontinuity in the I and Q plots 15ms after the start of the interruption. This discontinuity in the reflected signal is caused by the discontinuous increase in the incident signal, not any sharp change in $\Gamma$. That said, we can observe that after power is restored, Q (which depends only on temperature) immediately trends back to zero, while I (which depends on both temperature and charge density) lags for ~7ms before starting to trend back to zero, thus demonstrating the lag of recombination time between the equilibrium charge density (which is also only a function of temperature) and the actual charge density.

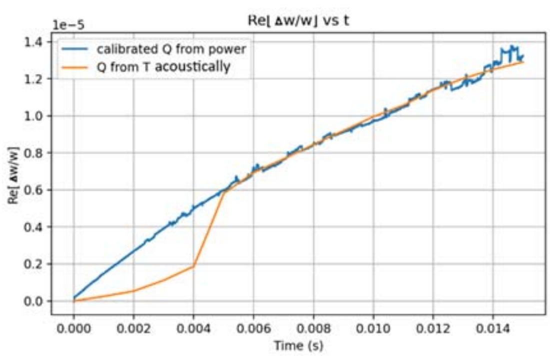


Fig. 6. A plot of Q measured from the reflected signal power aligned with a plot of Q derived from the acoustic temperature measurements. A scaling factor was chosen for the reflected power Q to maximize agreement with the acoustic Q.

Our measurements of I and Q are in volts and must be converted by a scaling factor before they can be used to measure the fractional change in the complex frequency. This scaling factor can be found experimentally by matching the measured Q to the Q predicted from acoustic temperature measurements. Figure 6. Shows that, when fitted with the appropriate scaling factor, we see strong agreement in the rates of change of these values (excluding the low confidence interval below 5ms). Figure 5.b then demonstrates that scaling factor applied to both I and Q for the time period of the interruption. With this scaling factor, we can now also use Q to derive a temperature as shown in Figure 2. Since these temperatures show strong agreement for interruptions longer than 5 ms, it is reasonable to assume that our Q-derived Temperature allows us to more reliably measure early time temperatures.

## Analysis of the Recombination Time

Using equations (5), our measurements for I, and our Q-derived temperature, we can evaluate the charge density as a function of time. Figure 7. shows this charge density as well as the equilibrium charge density predicted by Saha's Equation. Fitting exponentials to these plots yields a time constant of $\tau_s = 4.05\, ms$ for the equilibrium charge density and a time constant of $\tau_e = 11.46\, ms$ for the measured charge density.

The process for determining the time constant for how much the measured charge density lags behind the equilibrium is found in Landau's analysis of second viscosity. From it, we know that a quantity, $\xi$, of a fluid that is a function of density and temperature, with an equilibrium value $\xi_0$, will tend towards said equilibrium at a rate

$$\dot{\xi} = -(\xi - \xi_0)/\tau \qquad (7)$$

where $\tau$ is the time constant of the relaxation process. Allowing for $\xi$ to represent the actual charge density, $n_e$, and the equilibrium value with the charge density expected from Saha's equation, $n_{es} \approx n_e(0)e^{-t/\tau_s}$, we can find the recombination time, $\tau_r$, would then be found by solving

$$\dot{n_e} = -(n_e - n_e(0)e^{-t/\tau_s})/\tau_r \qquad (8)$$

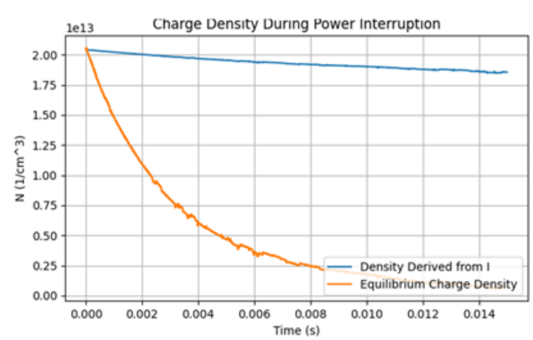


Fig. 7. Charge density and equilibrium charge density during a power interruption.

Which is a differential equation with the solution

$$n_e = n_e(0)e^{-t/\tau_r} + C\left[e^{-t/\tau_s} - e^{-t/\tau_r}\right] \qquad (9)$$

where $C = -n_e(0)\tau_s/(\tau_r - \tau_s)$. This solution can then be fitted to the observed charge density to find $\tau_r$. Doing so in yields a recombination time of 17 ms in our sulfur plasma.

We can now compare our experimental recombination time to the one predicted by Zeldovich's theory for three-body collisional recombination. For a collisionally dense but optically thin plasma like ours, the theory predicts that collisional recombination will be the dominant process and will have a frequency, $\nu$, of

$$\nu = \bar{v}_e \pi^2 r_0^5 n_e^2 \qquad (10)$$

where $\bar{v}_e$ is the thermal velocity and $r_0 = \frac{e^2}{6\pi\varepsilon_0 k_b T} \approx 3\ nm$ is the capture radius of the electrons. For a sulfur plasma running at a 4,000 K with a density of ~$10^{19}\ cc^{-3}$, this yields a predicted recombination time ($\tau_r = 1/\nu$) of 13 ms, which is remarkably close to the observed recombination time.

## Conclusion

Our experiment succeeded in measuring the recombination time of a dense plasma. In our sulfur bulb, we found it to be 17 ms. We verified our findings against Zeldovich three-body recombination theory, finding strong agreement.

Furthermore, we developed two new methods for measuring the instantaneous values of plasma properties, like temperature, charge density, and collision frequency, in real time. Our acoustic method can effectively measure plasma temperatures when the recombination time is substantially larger than the acoustic period and our reflection coefficient analysis likewise provides accurate measurements of these quantities up to an experimentally evaluated scaling factor. Because both of these methods found matching results when measuring the same quantity (temperature), we feel confident that they can be reused in future experiments of this nature.

Finding the recombination time of a plasma is vitally important to knowing the limiting time constraints on chemical or physical reactions that rely on rapidly shifting electron densities. For example, in the endeavor to create a self-oscillator as described in [1], we must have that the recombination time is far smaller than the acoustic resonance period. As we have shown, with a recombination time of 17 ms and an acoustic resonant frequency of ~30 kHz, this is clearly not possible in Sulfur. If a self-oscillator is possible to create with our system, it will need to be done with a different gas.

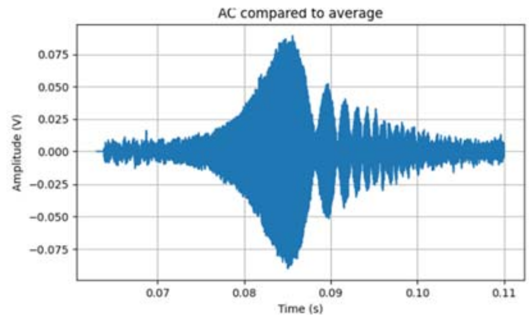


Fig. 9. A zoomed-in plot of the resonant oscillations in the AC photodiode of the plasma recovering from a 12 ms power interruption. The drive frequency is set above the threshold.

## Appendix: Resonant Oscillations of a Heating Driven Plasma

Our photodiode measurements show that, for drive frequencies below the quiescent plasma acoustic resonance but above some threshold minimum, oscillations like those in Figure 9. can be observed as the plasma heats back up to its quiescent temperature. These oscillations have a frequency matching the drive, increase in magnitude to some peak value, then decrease in magnitude while exhibiting beat-like behavior. This can be modeled as a damped, driven oscillator with a resonant frequency that is increasing in time.

$$\ddot{x} + \lambda\dot{x} + (\omega_0 + a * t)x = A\cos\big((\omega_D)t\big) \qquad (11)$$

Here y is the amplitude of the oscillation, $\lambda$ is the damping, A is the drive amplitude, $\omega_0$ and $\omega_D$are the resonant and drive frequencies respectively ($\omega_D > \omega_0$), and $a$ is the rate at which the resonance frequency is increasing in time (assumed linearly for simplicity).

Solving this numerically with parameters matching our experiment and an $a$ that is fitted by least squares (Figure 10.), yields a solution that bears a striking resemblance to the observed oscillations. Therefore, by fitting a solution, we can approximate the rate at which the resonance frequency of the plasma, and therefore its temperature is changing. Furthermore, for $\omega_D < \omega_0$, the solution produces oscillations that rapidly diminish, thus we can identify the threshold minimum drive frequency as where $\omega_D = \omega_0$, thus finding the lowest resonance (and temperature) attained by the plasma.

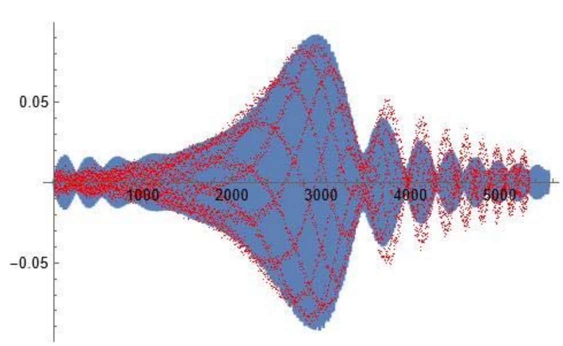


Fig. 10. The solution to equation (11) with best fitting parameters (blue) overlayed on top of the signal from Fig. 9. (red)

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